



US012418503B2

(12) **United States Patent**
Sen et al.

(10) **Patent No.:** **US 12,418,503 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **MANAGEMENT OF QUERIES IN
ELECTRONIC MAIL MESSAGES**

(71) Applicant: **Citigroup Global Markets Inc.**, New
York, NY (US)

(72) Inventors: **Deepesh Sen**, Princeton, NJ (US); **Itzel
Herrera**, Nutley, NJ (US); **Sumantha
Raghurama Shetty**, Pune (IN);
Prashant Rajendrakumar Shah, Pune
(IN); **Steve Morgan**, Cliffside Park, NJ
(US)

(73) Assignee: **Citigroup Global Markets Inc.**, New
York, NY (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/374,189**

(22) Filed: **Sep. 28, 2023**

(65) **Prior Publication Data**
US 2024/0267355 A1 Aug. 8, 2024

Related U.S. Application Data

(63) Continuation of application No. 18/107,911, filed on
Feb. 9, 2023, now Pat. No. 11,777,892, which is a
continuation of application No. 18/107,368, filed on
Feb. 8, 2023, now Pat. No. 11,777,886.

(51) **Int. Cl.**
H04L 51/42 (2022.01)
G06Q 10/107 (2023.01)
G06Q 10/1093 (2023.01)
H04L 51/08 (2022.01)
H04L 51/216 (2022.01)
H04L 51/48 (2022.01)

(52) **U.S. Cl.**

CPC **H04L 51/42** (2022.05); **G06Q 10/107**
(2013.01); **G06Q 10/1095** (2013.01); **H04L**
51/08 (2013.01); **H04L 51/216** (2022.05);
H04L 51/48 (2022.05)

(58) **Field of Classification Search**

None
See application file for complete search history.

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709/248

(Continued)

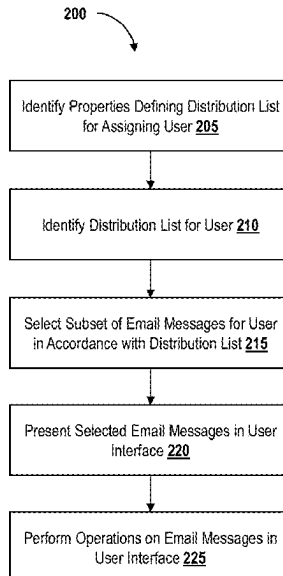
Primary Examiner — Ranodhi Serrao

(74) *Attorney, Agent, or Firm* — Foley & Lardner LLP

(57) **ABSTRACT**

Presented herein are systems and methods for handling
email messages. An application may identify, via the user
interface, one or more properties defining a distribution list
to assign a user of the client device. The application may
transmit, to the server, the one or more properties to assign
the user of the client device to the distribution list, with
which at least one of a plurality of email messages to is
selected for sending to the client device. The application
may receive, from the server, an identification of a subset of
email messages selected from the plurality of email mes-
sages based on an assignment of the user of the client device
to the distribution list and content in the subset of email
messages. The application may present, in the user interface,
a message list of the subset of email messages.

20 Claims, 18 Drawing Sheets



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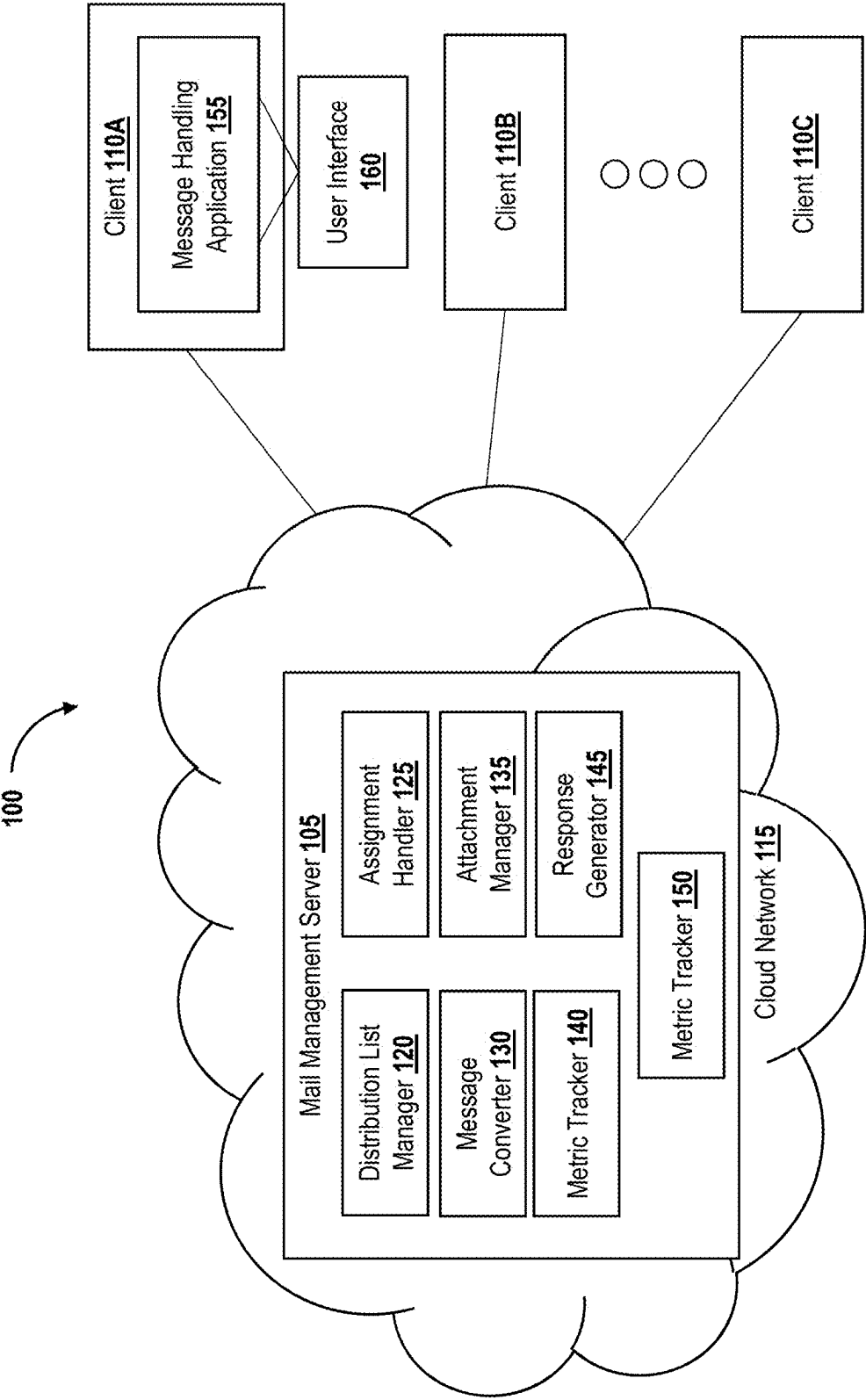


FIG. 1

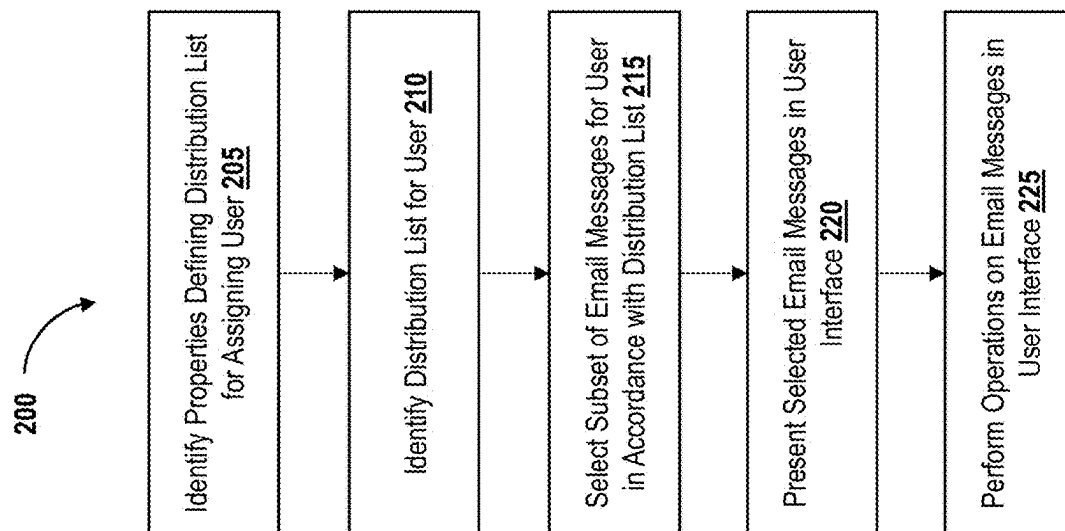


FIG. 2

300

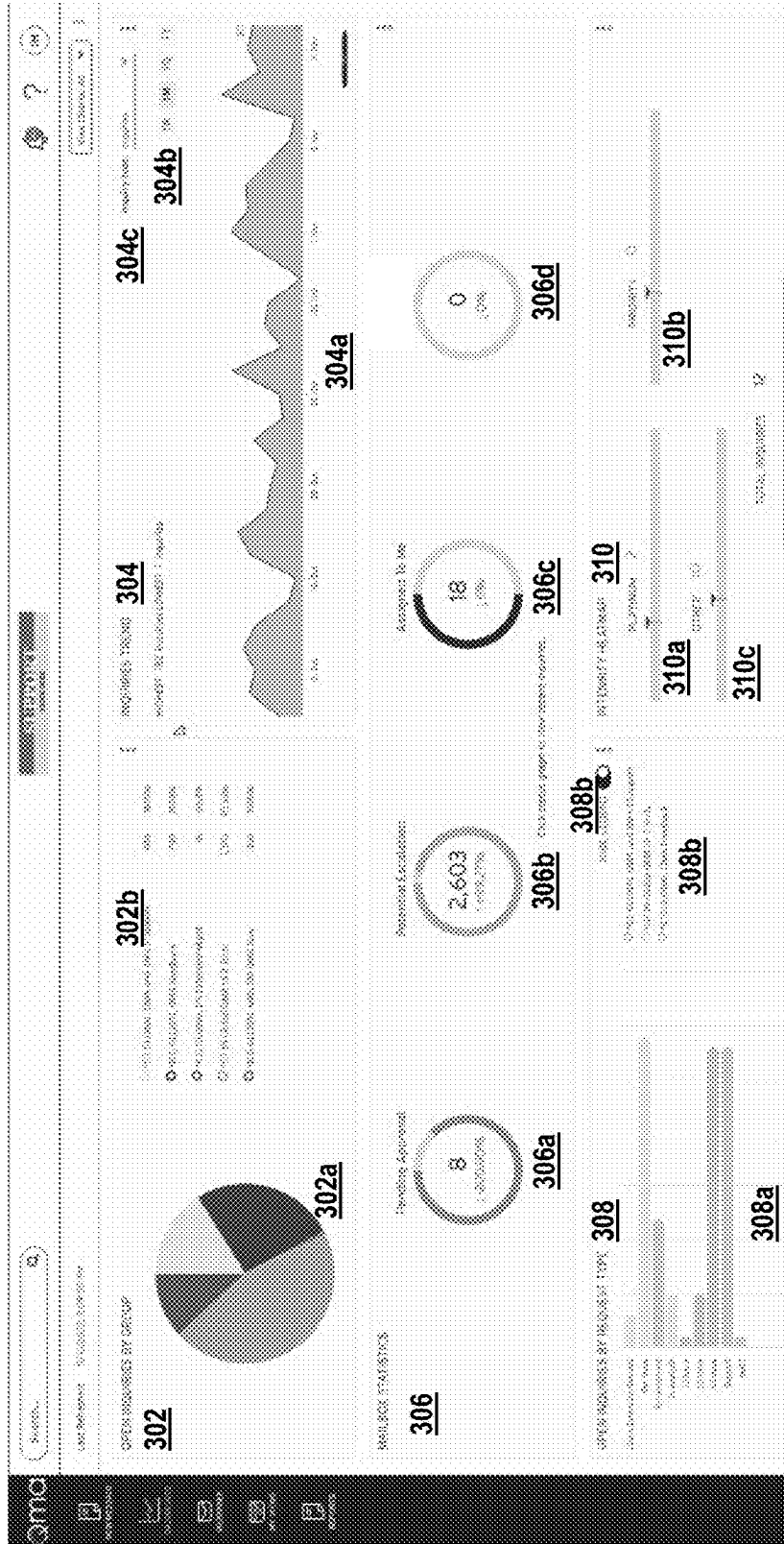


FIG. 3

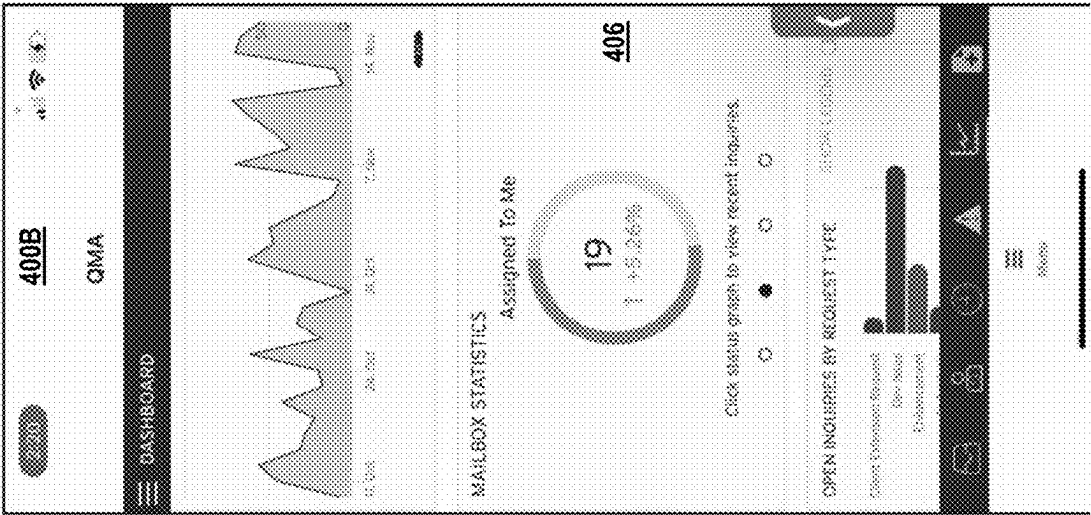


FIG. 4B

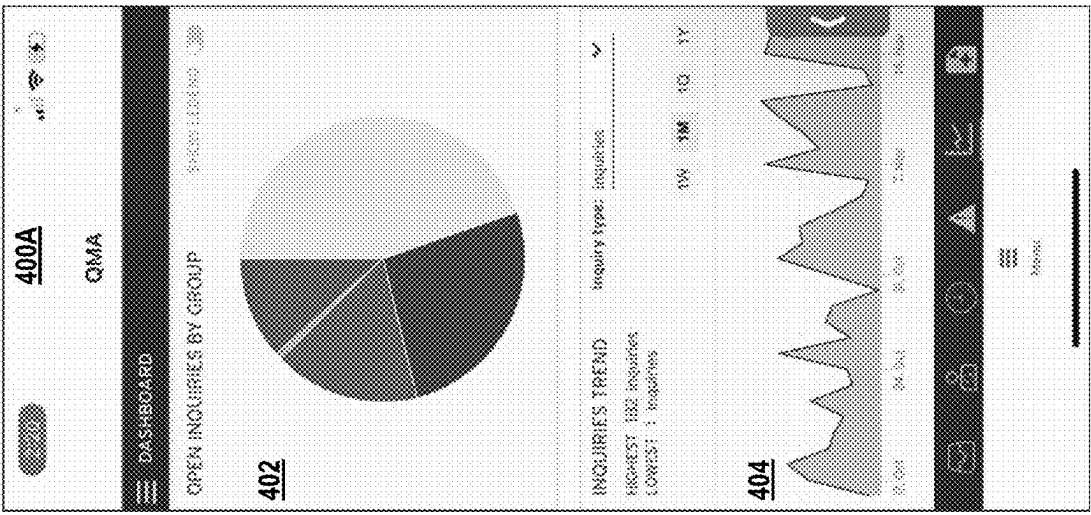


FIG. 4A

500

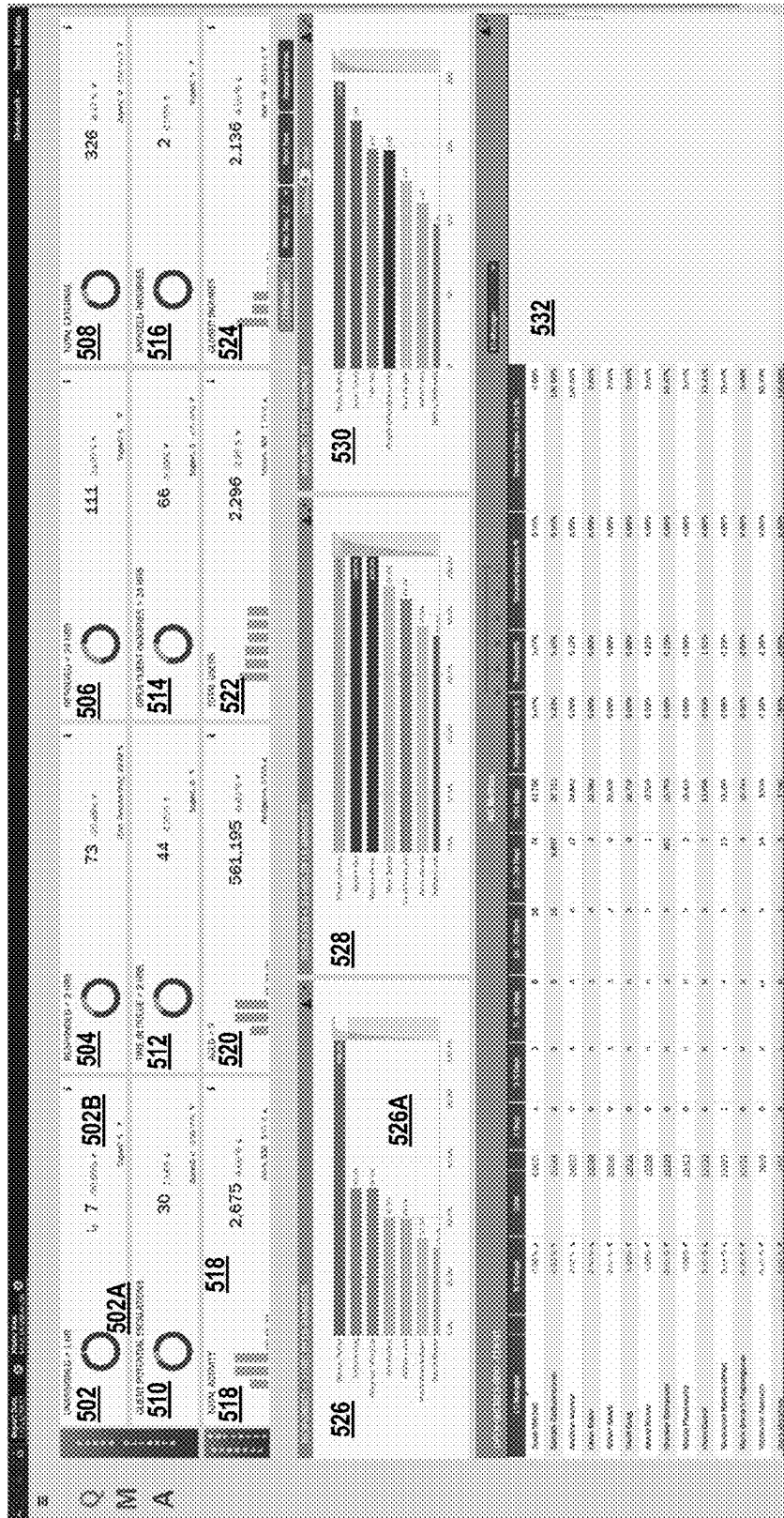


FIG. 5

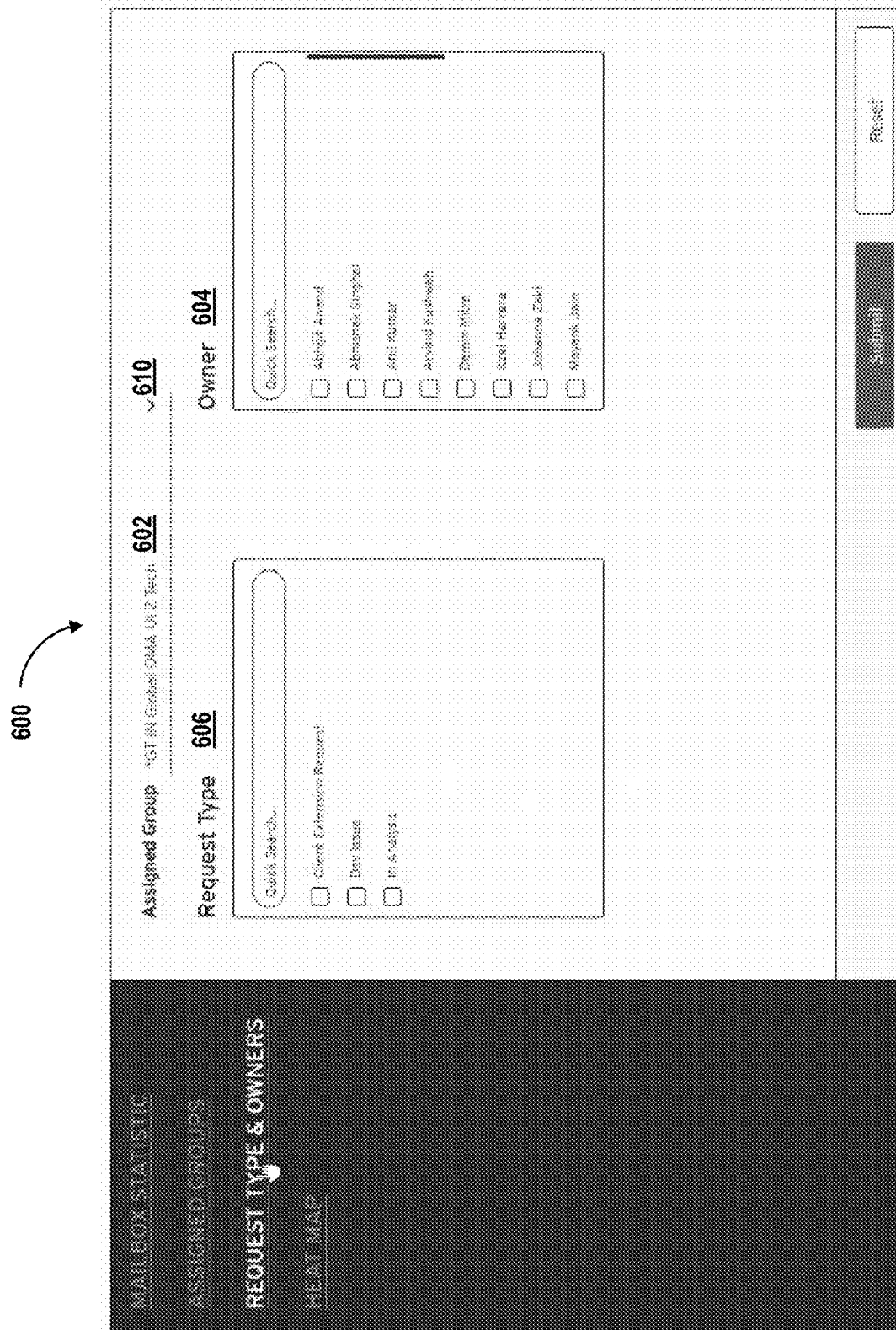


FIG. 6

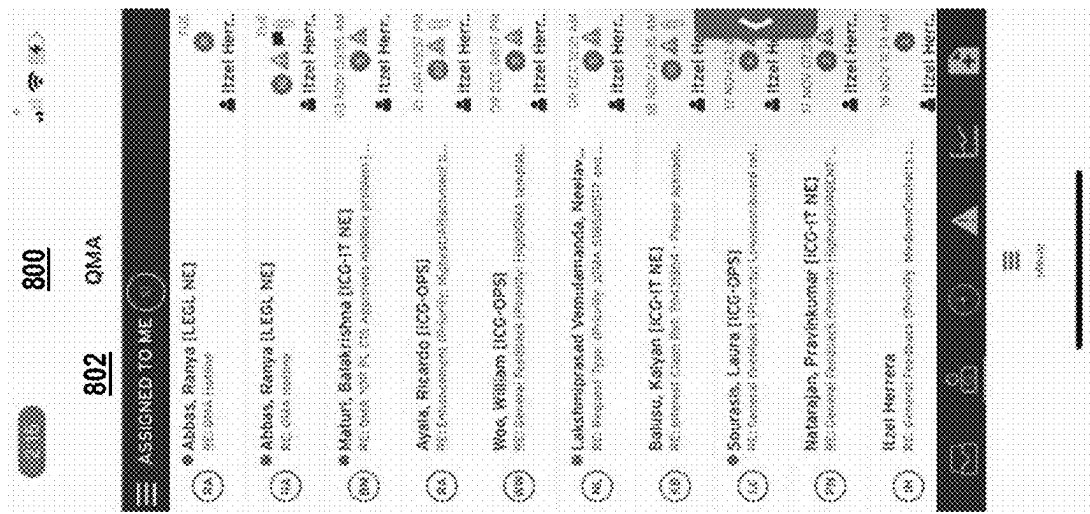


FIG. 8

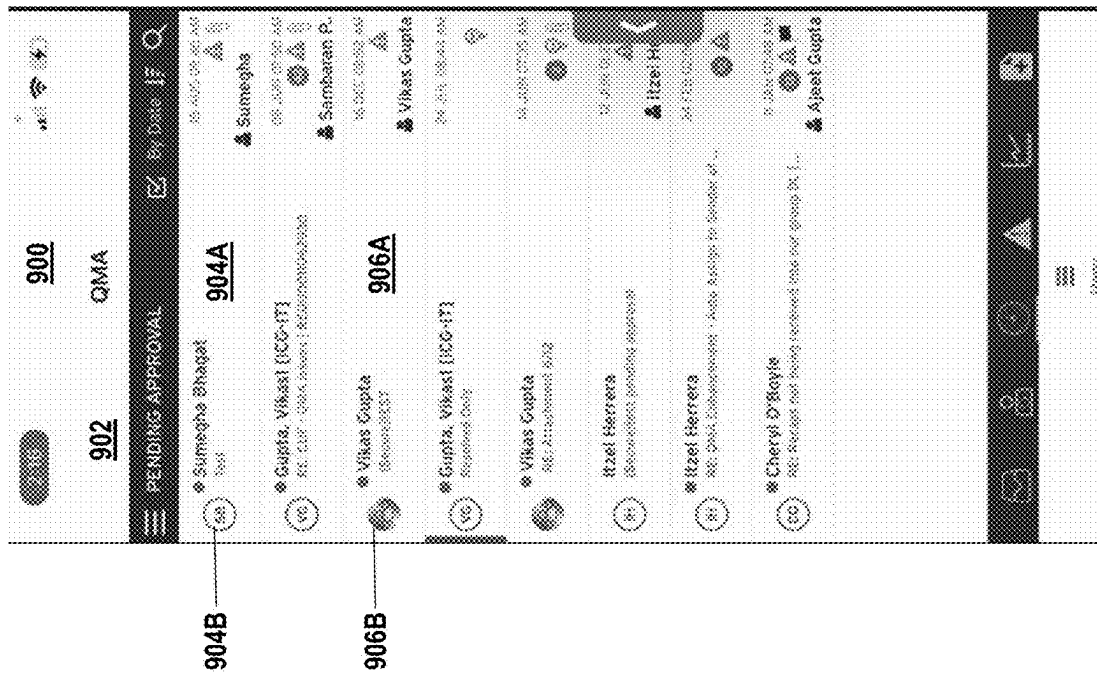


FIG. 9

1000

1002

QMA

POTENTIAL ESCALAT...

QMA

66

Bobek, Agnieszka [CG-OPS]

RC: 2023/02/02/20

FIG. 10

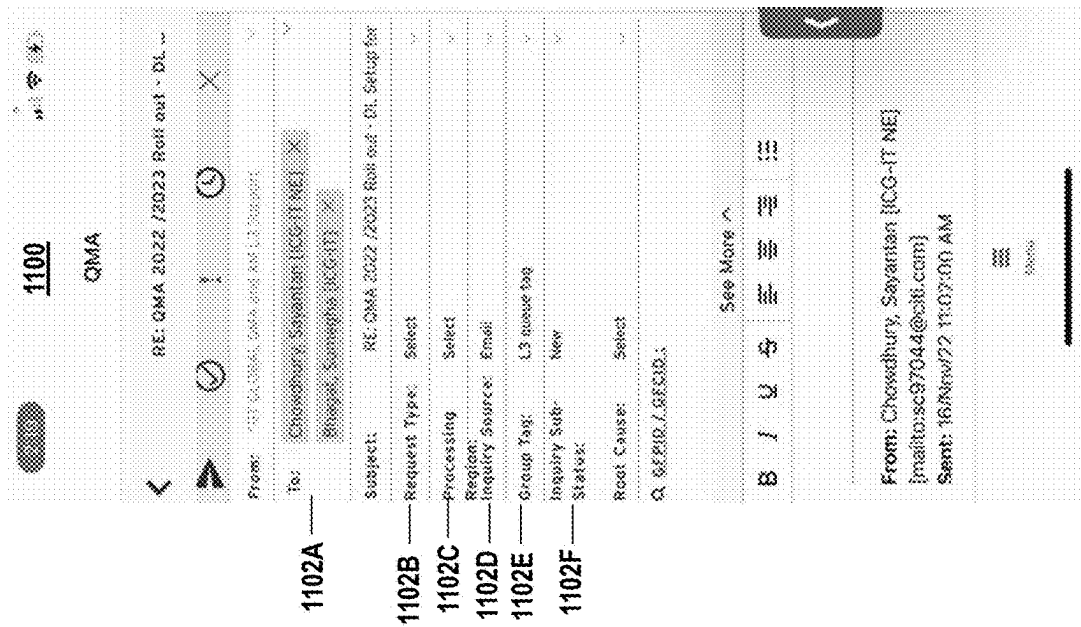


FIG. 11

1200

×

RESOLVE INQUIRY

From :

400 GLOBAL QMA Feedback

1202

Request type:

Request:

1204

Sub-Status :

Completed

1206

☐ Resolve for all my groups

☒ Actioned --No Reply

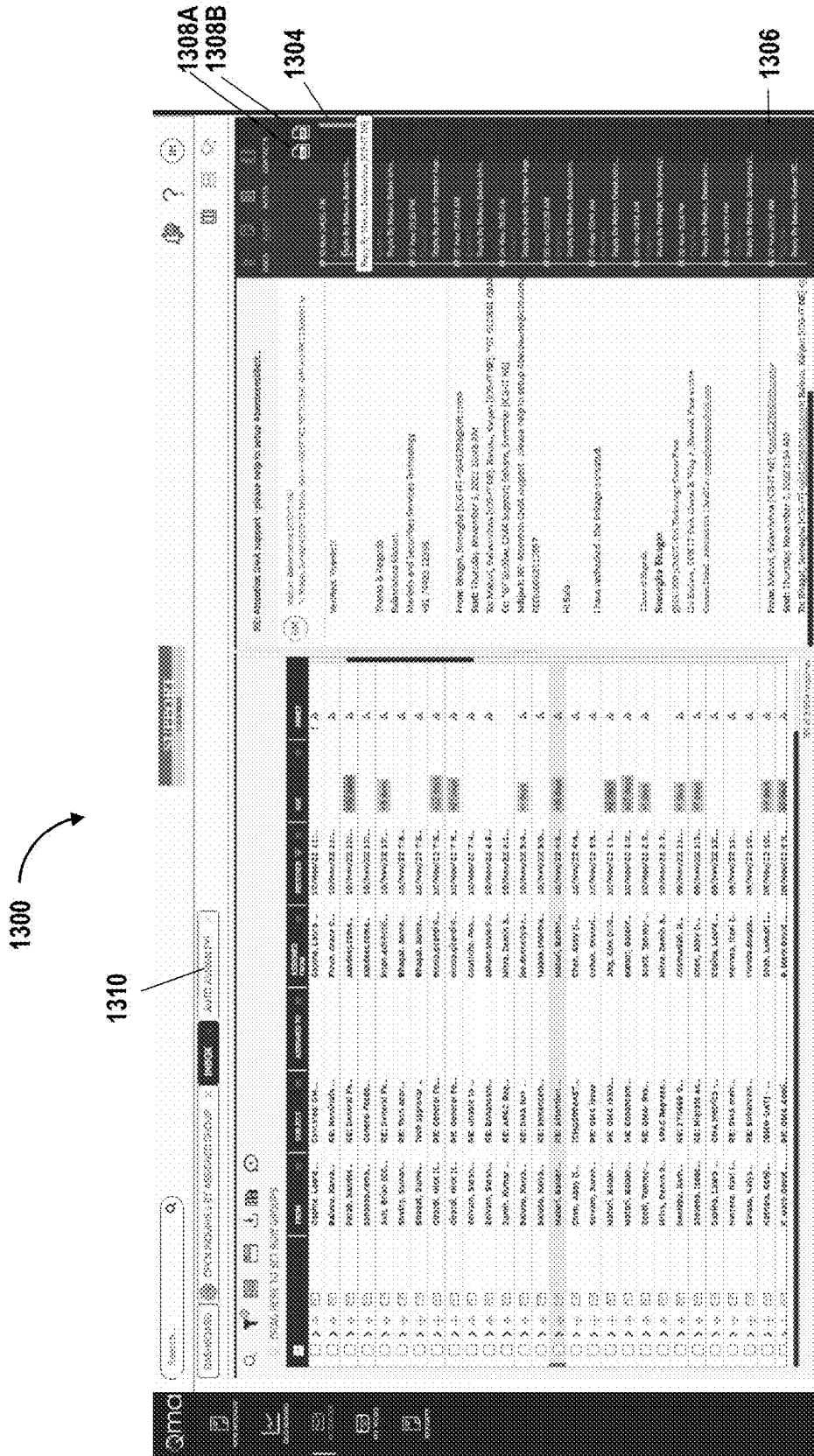
1208

More Fields

Submit

Reset

FIG. 12



1400

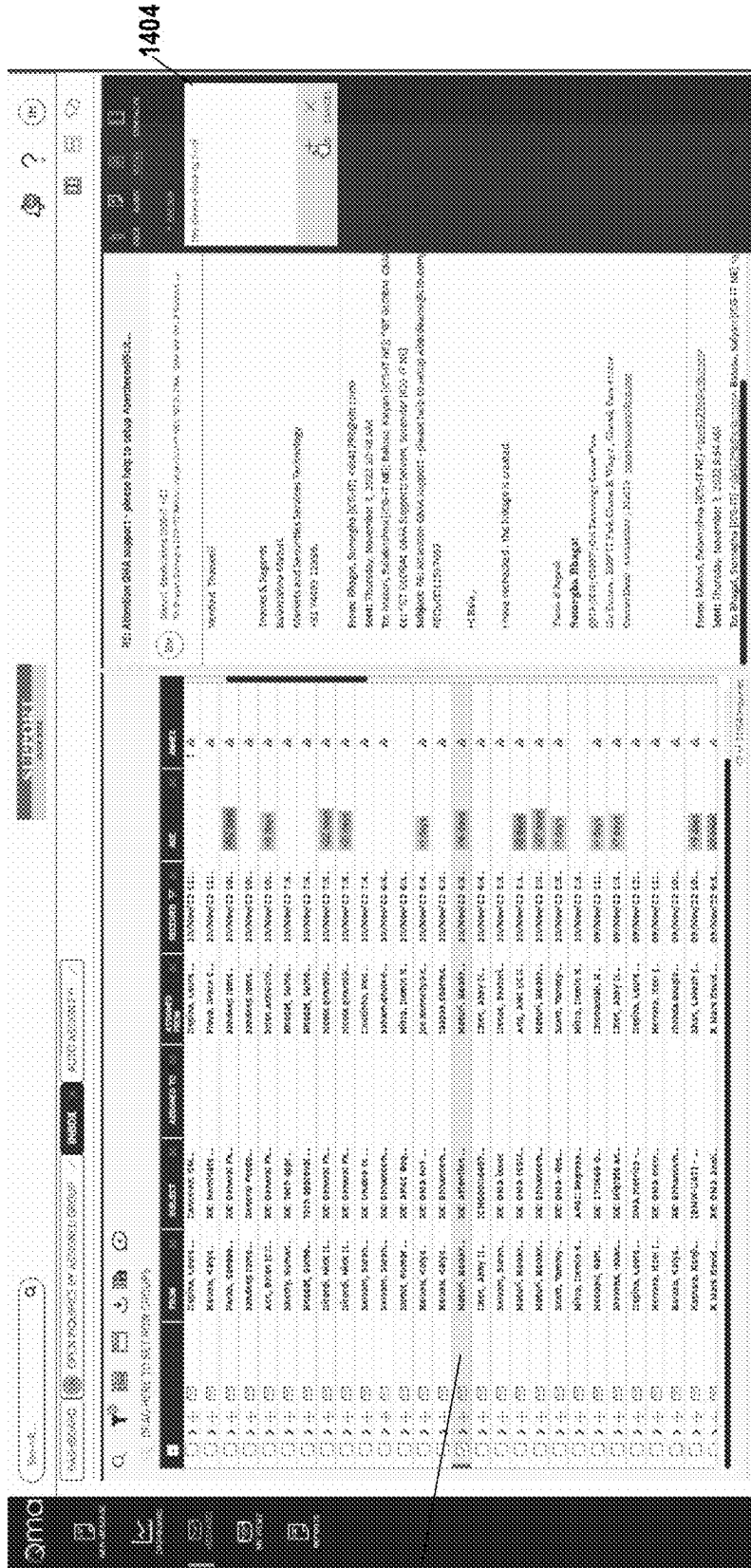


FIG. 14

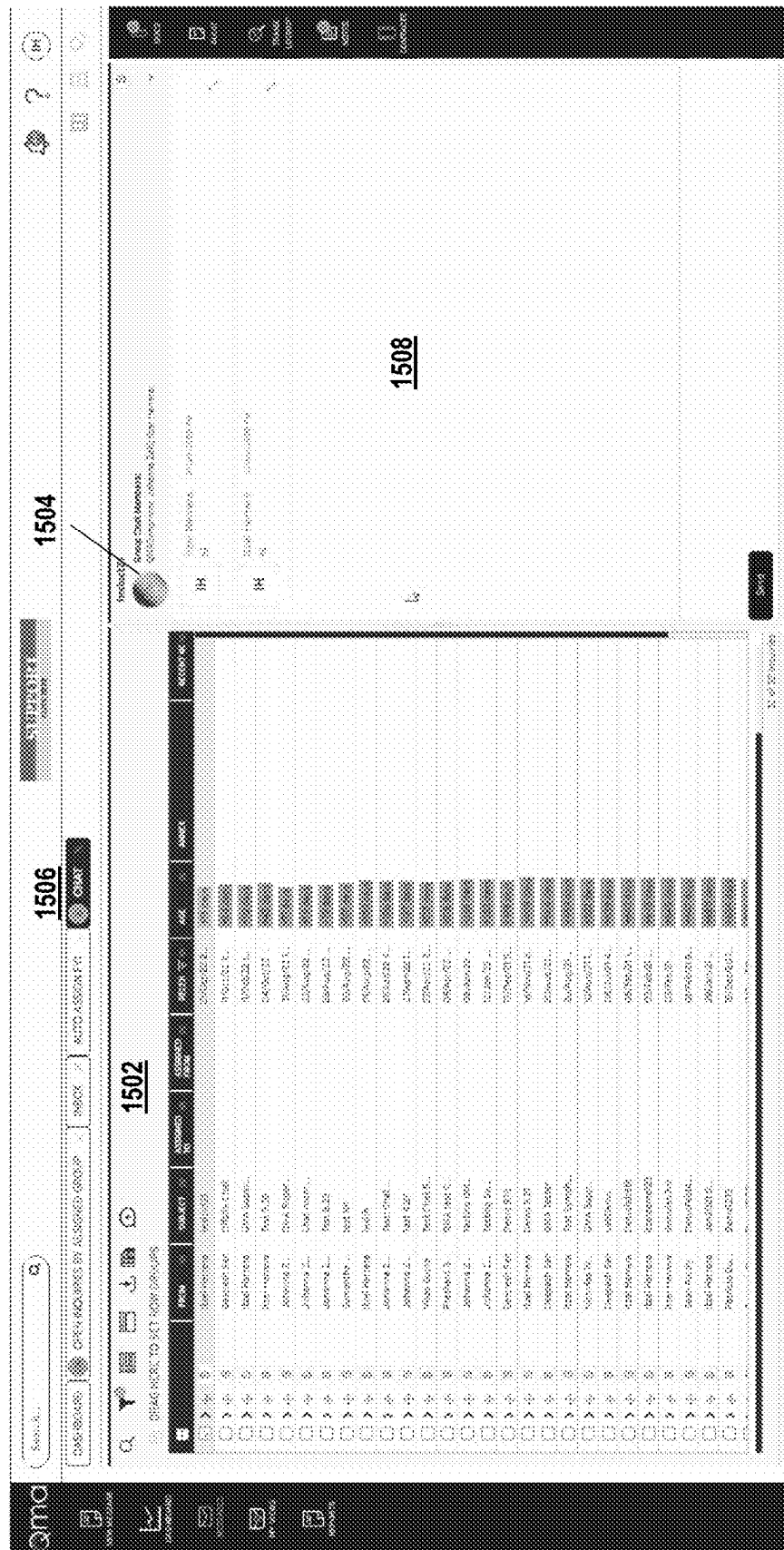


FIG. 15

1600A

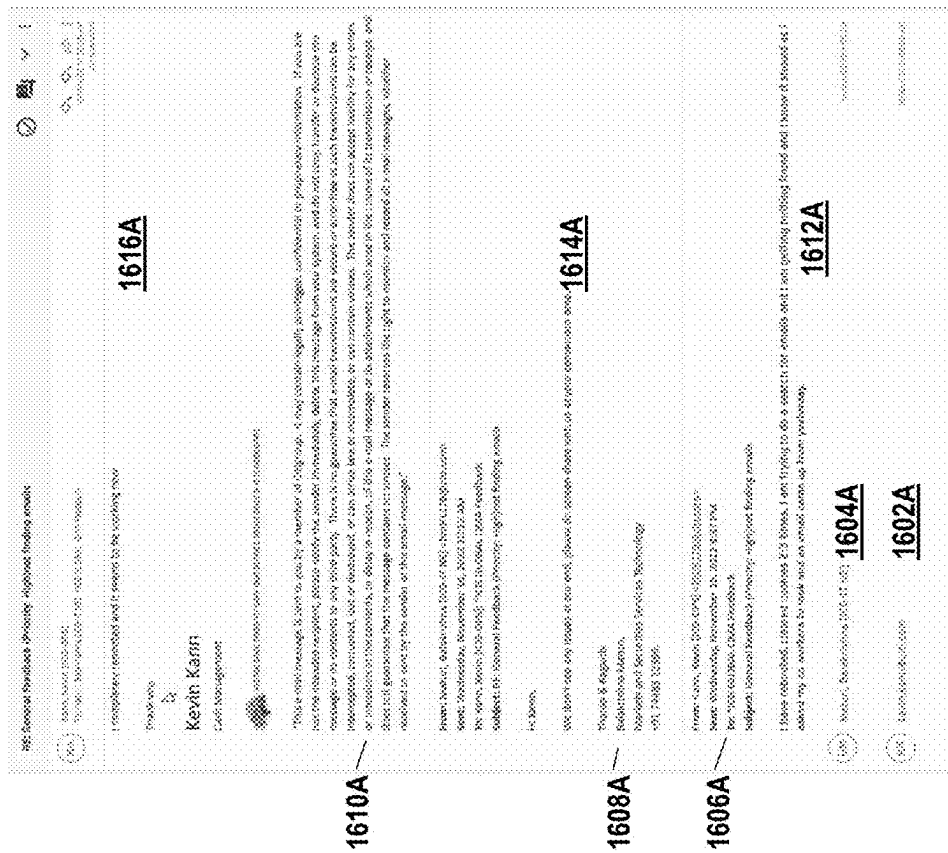


FIG. 16A

1600B



FIG. 16B



FIG. 17

1800

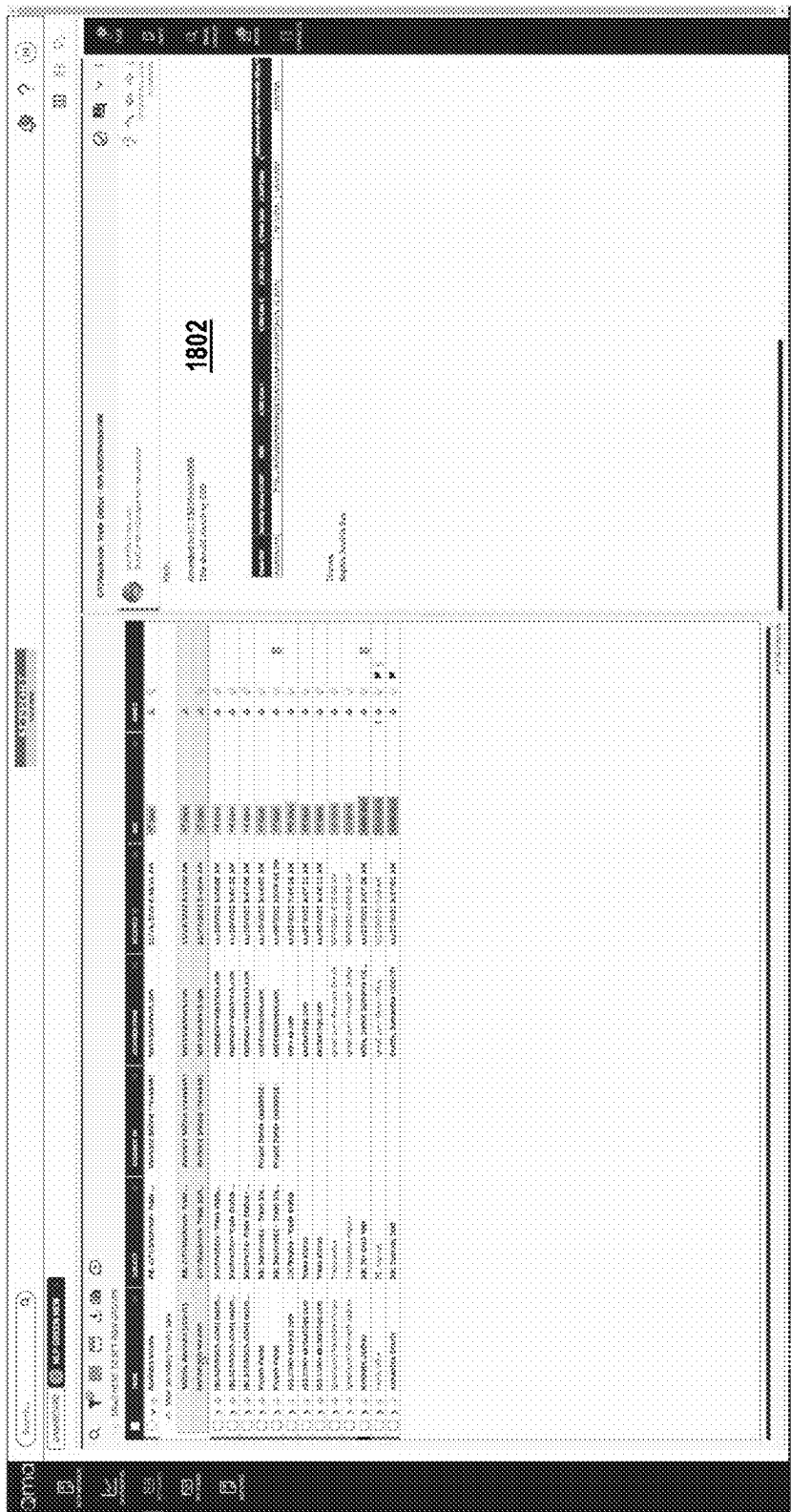


FIG. 18

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**MANAGEMENT OF QUERIES IN
ELECTRONIC MAIL MESSAGES****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application claims the benefit of priority under 35 U.S.C. § 120 as a continuation of U.S. patent application Ser. No. 18/107,911, filed Feb. 9, 2023, which claims the benefit of priority under 35 U.S.C. § 120 as a continuation of U.S. patent application Ser. No. 18/107,368, filed Feb. 8, 2023, each of which is incorporated by reference in their entireties.

TECHNICAL FIELD

This application generally relates to communications. In particular, the present application relates to managing and handling electronic mail messages.

BACKGROUND

A mail agent running on a computing device may be used by users to send and receive electronic mail messages with one another. A mail server may facilitate the transfer of electronic mail messages from one user to another user. When an electronic mail message identifies a particular user as a recipient, the mail server may forward the electronic mail message to the recipient user. Upon receipt, the electronic mail messages may be displayed in an inbox user interface of the mail agent. The inbox user interface can organize the electronic mail messages as such messages received. As more and more electronic mail messages are amassed by the recipient, the recipient's inbox may become cluttered and disorderly, resulting in a degradation in the quality of human computer interactions (HCI) between the user and the user interface of the mail agent. Without any means to organize the received messages, certain electronic mail messages may be missed by the user.

SUMMARY

Presented herein are systems and methods for managing and handling electronic mail messages. A sender may use a mail agent or application running on a client device to compose and send an electronic mail message to another user. The electronic mail message may include headers (e.g., identifying a recipient, sender, and subject), a body message, and one or more tags, among others. A mail server interfacing with the mail agent on the client device can receive and direct the electronic mail message using the recipient identified in the header of the electronic mail message. The recipient client device may receive the electronic mail message from the sender for display via a user interface for the recipient user. The user interface of the mail agent may include an inbox to display a list of received electronic mail messages and a message window to show

Although the recipient user may use the mail agent to access and read received electronic mail messages, utility of the user interface of the mail agent may degrade as more and more electronic mail messages accumulate in the user's inbox. This is because the sheer amount of electronic mail messages may become cluttered and disorderly, especially without additional organization of such messages. Such a cluttered inbox may lead to the user becoming frustrated with the mail agent, resulting in the degradation of the quality of human-computer interactions (HCI) between the user and the user interface of the agent. The disarrangement

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of the user interface of the mail agent may also lead to higher consumption of computing resources and more time, because the user may have to submit multiple search requests to find desired messages.

One attempt to addressing some of these problems may be to set up rules to classify emails based on header or body content of received electronic mail messages. This approach, however, may be cumbersome and tedious, because the user may have to manually set up such rules, defining individual keywords, senders, or recipients. Moreover, manually setting up rules may also provide limited and inflexible controls over the organization of the received electronic mail messages. These controls may result in over-inclusion of unwanted electronic mail messages and under-inclusion of desired electronic mail messages, still leaving much to be desired in the quality of HCI between the user and the mail agent.

To account for these and other technical problems, distribution lists assigned to various users may be used to identify electronic mail messages to bring to the recipient's attention. A mail handling application (also referred herein as a mail user agent) on a client device may provide a dialog box to select parameters defining which distribution list the user is to be assigned. The parameters may include, for example, a request type, a group identifier, and other information about the user. Upon selection, the mail handling application may send the parameters to the mail server. The mail server may receive the parameters defining the distribution list from the client device. Using the parameters, the mail server may identify the distribution list to which to assign the user.

With the assignment, whenever an electronic mail message is received, the mail server may check the distribution list for the recipient user. Using the distribution list, the mail server may parse the contents (e.g., header, body message, and tags) of the electronic mail message to determine whether the electronic mail message is to be assigned to the user. If the contents of the message is determined to correspond to the distribution list, the mail server may identify the electronic mail message as to be assigned. The mail server may send the message with an identification that the message is assigned to the user. For instance, when the content of the message corresponds to a request type for the distribution list, the mail server may determine that the message is to be brought to the attention of the user. On the other hand, if the contents of the message is determined to not correspond to the distribution list, the mail server may identify the electronic mail message as to be not assigned to the user. The mail server may forward the electronic mail message to the user, without any identification. Upon receipt, the mail handling application may present the electronic mail message in an inbox dedicated for messages selected using the distribution list to bring such messages to the user's attention.

In addition, the mail handling application in conjunction with the mail server may provide other various functionalities for administering the presentation of the electronic mail messages on the user interface. For example, from the list of electronic mail messages, the user of the application may identify a message to re-assign, and the application may send a request to change assignment to another user. The application (or the server) may also determine whether an electronic mail message that has been assigned to the user has been replied to within a set time window. If the set time window has passed without any replies by the user, the application may provide an escalation indication that the message has not been responded to. Furthermore, the appli-

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cation may also send an identification of the message to another user (e.g., a manager of the initial employee assigned to handling the email).

Upon request to convert electronic mail messages into online chat conversation, the application (or the server) may generate an online chat conversation by extracting main content in body message and removing auxiliary content (e.g., signatures, disclaimers, etc.) from a thread of messages. Conversely, upon receiving a request to convert an online chat conversation to an electronic mail thread, the application may interface with a chat application to retrieve the chat conversation thread. With the identification, the application may generate the electronic mail thread by adding electronic mail message formatting (e.g., headers, subject, and signature) to each of the chat conversations. Furthermore, the application may administer a user's calendar. Whenever a request to cancel an event is received, the application may identify another calendar event for the time slot as the canceled event, and may present the identified calendar event as a recommendation to the user. The application may also provide automatically generated responses for a query in an electronic mail message upon request.

In this manner, the mail handling application in combination with the mail server may improve the quality human-computer interaction (HCI) between the user and the electronic mail messages by providing an ability to manage arrangement of such messages. For example, use of the distribution list to select which electronic mail message to bring to the user's attention may reduce the cluttering and disorderliness of an inbox of the application, permitting the user to access relevant received messages and ability to quickly reply to such messages. The other capabilities of the user interface may further organize the electronic mail messages in the user interface of the mail handling application to improve the quality of HCI between the user and the application. These may also further reduce consumption of computing resources, from eliminating additional interactions to achieve similar results.

Aspects of present disclosure are directed to systems, methods, and non-transitory computer readable media for handling electronic mail messages in networked environments. The system may include a client device having one or more processors coupled with memory in communication with a server. The system may an application executable on the client device. The application may have a user interface for handling electronic mail messages. The application may identify, via the user interface, one or more properties defining a distribution list to assign a user of the client device. The application may transmit, to the server, the one or more properties to assign the user of the client device to the distribution list, with which at least one of a plurality of electronic mail messages to is selected for sending to the client device. The application may receive, from the server, an identification of a subset of electronic mail messages selected from the plurality of electronic mail messages based on an assignment of the user of the client device to the distribution list and content in the subset of electronic mail messages. The application may present, in the user interface, a message list of the subset of electronic mail messages.

In one embodiment, the application may receive, via the user interface, a request to present an electronic mail thread corresponding to a second subset of electronic mail messages, as a chat conversation thread. The application may identify, from each of the second subset of electronic mail messages, main content and auxiliary content, responsive to the request to present. The application may generate the chat conversation thread including the main content identified

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from each of the second subset of electronic mail messages, while excluding the auxiliary content. The application may present, in a message panel of the user interface, at least a portion of the chat conversation thread.

In another embodiment, the application may receive, via the user interface, a request to add a chat conversation thread from a second application as one of the subset of messages in the message list. The application may generate an electronic mail thread to include a second subset of messages corresponding to a plurality of conversations of the chat conversation thread. The application may present, via the message list, an identification of the electronic mail thread among a plurality of threads.

In yet another embodiment, the application may identify, from the subset of electronic mail messages, an electronic mail message for no response is sent within a defined time window after receipt. The application may provide, via the user interface, an indicator identifying the electronic mail message as having no response within the defined time window.

In yet another embodiment, the application may receive, via the user interface, a request to change assignment of an electronic mail message of the subset of electronic mail messages to another user. The application may transmit, to the server, an indication that the electronic mail message is to be assigned to the second distribution list, responsive to the request to change assignment of the electronic mail message. The application may remove the electronic mail message among the subset of electronic mail messages from presentation in the message list.

In yet another embodiment, the application may receive, via the user interface, a request to cancel a first calendar event corresponding to one of the subset of electronic mail messages. The application may identify, responsive to the request to cancel, a second calendar event for a time slot at least partially concurrent with the first calendar event. The application may present, via the user interface, an identification of the second calendar event.

In yet another embodiment, the application may provide, for at least one message of the subset of electronic mail messages, a response based on at least a portion of the at least one message. In yet another embodiment, the application may present, in a dashboard of the user interface, a statistic indicator associated with provision of the plurality of electronic mail messages to the client device in accordance with the distribution list.

In yet another embodiment, the application may provide, via an attachment window at least partially concurrently with the message list of the user interface, a plurality of attachments available to be added to at least one of the subset of messages. In yet another embodiment, the client device may include a mobile communication device in communication with the server on a cloud computing network. The user interface may have a size corresponding to a display of the mobile communication device.

Aspects of present disclosure are directed to systems, methods, and non-transitory computer readable media for managing electronic mail messages in networked environments. The system may include at least one server having one or more processors coupled with memory in communication with a plurality of client devices. The at least one server may receive one or more properties defining a distribution list to assign a user associated with a client device of the plurality of client devices. The at least one server may identify, from a plurality of distribution lists, the distribution list to which to assign the user based on the one or more properties. The at least one server may select, from a

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plurality of electronic mail messages to be sent to the client device, a subset of electronic mail messages based on assignment of the user to the distribution list and content in the subset of electronic mail messages. The at least one server may transmit, to the client device, an identification of the subset of electronic mail messages for presentation in a message list of a user interface.

In one embodiment, the at least one server may receive, from the client device, a request to present an electronic mail thread corresponding to a second subset of electronic mail messages, as a chat conversation thread. The at least one server may identify, from each of the second subset of electronic mail messages, main content and auxiliary content, responsive to the request to present. The at least one server may generate the chat conversation thread including the main content identified from each of the second subset of electronic mail messages, while excluding the auxiliary content. The at least one server may send, to the client device, the chat conversation thread for presentation in a message panel of the user interface.

In another embodiment, the at least one server may receive, from the client device, a request to add a chat conversation thread from a second application as one of the subset of messages in the message list. The at least one server may generate an electronic mail thread to include a second subset of messages corresponding to a plurality of conversations of the chat conversation thread. The at least one server may send, to the client device, an identification of the electronic mail thread among a plurality of threads for presentation in the message list.

In yet another embodiment, the at least one server may identify, from the subset of electronic mail messages, an electronic mail message for no response is sent within a defined time window after receipt. The at least one server may send, to the client device, an indicator identifying the electronic mail message as having no response within the defined time window via the user interface.

In yet another embodiment, the at least one server may receive, from the client device, a request to change assignment of an electronic mail message of the subset of electronic mail messages to another user. The at least one server may identify, from a plurality of users, a second user to which to assign the electronic mail message based on the request to change. The at least one server may send, to a second client device associated with the second user, an identification of the electronic mail message.

In yet another embodiment, the at least one server may receive, from the client device, a request to cancel a first calendar event corresponding to one of the subset of electronic mail messages. The at least one server may identify, responsive to the request to cancel, a second calendar event for a time slot at least partially concurrent with the first calendar event for the user. The at least one server may send, to the client device, an identification of the second calendar event for the user.

In yet another embodiment, the at least one server may select, from a plurality of distribution lists, the distribution list for the user based on the one or more properties received from the client device. In yet another embodiment, the at least one server may generate, for at least one message of the subset of electronic mail messages, a response based on at least a portion of the at least one message. In yet another embodiment, the at least one server may comprise at least one of a mail exchange server or a dedicated server for interfacing with an application for handling messages executing on the client device. In yet another embodiment,

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the at least one server may be distributed in a cloud computing network in communication with the plurality of client devices

It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the embodiments described herein.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings constitute a part of this specification, illustrate an embodiment, and together with the specification, explain the subject matter of the disclosure.

FIG. 1 depicts a block diagram of an example system for managing and handling electronic mail messages in accordance with an illustrative embodiment;

FIG. 2 depicts a flow diagram of an example method of managing and handling electronic mail messages in accordance with an illustrative embodiment;

FIG. 3 depicts a screenshot of a dashboard of mailbox statistics displayed via an interface in accordance with an embodiment;

FIGS. 4A and 4B each depict a screenshot of a dashboard of mailbox statistics displayed via an interface for a mobile device, in accordance with an illustrative embodiment;

FIG. 5 depicts a screenshot of a window of various statistics displayed via an interface for a mobile device, in accordance with an illustrative embodiment;

FIG. 6 depicts a screenshot of a dialogue box for selecting parameters to define a distribution list to which to assign a user in accordance with an illustrative embodiment;

FIG. 7 depicts a screenshot of an interface of an application including a message list and a message panel in accordance with an illustrative embodiment;

FIG. 8 depicts a screenshot of an interface of an application for a mobile device including a message list in accordance with an illustrative embodiment;

FIG. 9 depicts a screenshot of an interface of an application for a mobile device including a message list with indicators regarding approval in accordance with an illustrative embodiment;

FIG. 10 depicts a screenshot of an interface of an application for a mobile device including a message list with indicators for potential escalations in accordance with an illustrative embodiment;

FIG. 11 depicts a screenshot of an interface of an application for a mobile device including a message with various tags in accordance with an illustrative embodiment;

FIG. 12 depicts a screenshot of a dialogue box for an inquiry resolution in accordance with an illustrative embodiment;

FIG. 13 depicts a screenshot of an interface of an application including a message list and a message panel with a side panel including an audit of events in accordance with an illustrative embodiment;

FIG. 14 depicts a screenshot of an interface of an application including a message list and a message panel with a side panel including notes in accordance with an illustrative embodiment;

FIG. 15 depicts a screenshot an interface of an application including a message list and a message panel with an electronic mail thread converted to a chat conversation thread, in accordance with an illustrative embodiment;

FIG. 16A depicts a screenshot of a selected electronic mail thread with multiple messages in accordance with an illustrative embodiment;

FIG. 16B depicts a screenshot of a chat conversation thread converted from the electronic mail thread in accordance with an illustrative embodiment;

FIG. 17 depicts a screenshot of a dialogue box for suggested automated responses in accordance with an illustrative embodiment; and

FIG. 18 depicts a screenshot of an automated reply included in a response electronic mail message in accordance with an embodiment.

DETAILED DESCRIPTION

Reference will now be made to the embodiments illustrated in the drawings, and specific language will be used here to describe the same. It will nevertheless be understood that no limitation of the scope of the disclosure is thereby intended. Alterations and further modifications of the features illustrated here, and additional applications of the principles as illustrated here, which would occur to a person skilled in the relevant art and having possession of this disclosure, are to be considered within the scope of the disclosure.

The present disclosure is directed to systems and methods for managing and handling electronic mail messages. Users of an electronic mail message service may be assigned to one or more distribution lists based on parameters configured by the users. The distribution lists may include one or more users (recipients), and the message may be intended for one or more of the users. The parameters may include, for example, a request type, a group identifier, and other information about the user. Once assigned, the server may determine that an electronic mail message directed at the user is to be brought to the user's attention based on the contents of the message as well as the distribution list to which the user belongs. From the determination, the server may provide an identification that the electronic mail message is assigned to the user to a client device associated with the user. Upon receipt, a mail management application on the client device may include the identified message in a list of messages (e.g., inbox) assigned to the user. In this manner, the cluttering and disorderliness of the message list can be reduced, thereby improving human-computer interaction (HCI) between the user and the application and freeing up consumption of computing resources that would have been spent in additional interactions to access messages.

FIG. 1 depicts a block diagram of a system 100 for binding web components to protect accessing of resources. The system 100 may include at least one mail management server 105 and one or more clients 110A-N (hereinafter generally referred to as clients 110) communicatively coupled via at least one cloud network 115. The mail management server 105 may include at least one distribution list manager 120, at least one assignment manager 125, at least one message converter 130, at least one attachment manager 135, at least one calendar manager 140, at least one response generator 145, at least one metric tracker 150, among others. At least one client 110 may include at least one message handling application 155 to provide at least one user interface 160. Embodiments may comprise additional or alternative components or omit certain components from those of FIG. 1, and still fall within the scope of this disclosure.

Various hardware and software components of one or more public or private networks (e.g., over the cloud network 115) may interconnect the various components of the system 100. Non-limiting examples of such networks may

include Local Area Network (LAN), Wireless Local Area Network (WLAN), Metropolitan Area Network (MAN), Wide Area Network (WAN), and the Internet. The communication over the network may be performed in accordance with various communication protocols, such as Transmission Control Protocol and Internet Protocol (TCP/IP), User Datagram Protocol (UDP), and IEEE communication protocols, among others.

The cloud computing network 115 may include a cloud computing environment, with one or more nodes (e.g., one or more servers) can be distributed over multiple sites (e.g., data centers). The cloud computing network 115 may be used to implement or execute the mail management server 105 and its components, such as the distribution list manager 120, the assignment manager 125, the message converter 130, the attachment manager 135, the calendar manager 140, at least one response generator 145, the metric tracker 150, among others. For instance, the distribution list manager 120 and the assignment manager 125 may be executed on servers at one site, while the message converter 130, the attachment manager 135, the calendar manager 140, at least one response generator 145, the metric tracker 150 may be run out of other sites. The cloud computing network 115 may be setup in accordance with any number of service models, such as an infrastructure as a service (IaaS), a platform as a service (PaaS), or a software as a service (SaaS), among others.

The client 110 may be any computing device comprising one or more processors coupled with memory and software, and capable of performing the various processes and tasks described herein. The client 110 may be, for example, a desktop, a thin client, a laptop, a tablet, and a mobile phone, among others. The client 110 may be in communication with the mail management server 105, via the cloud network 115. Communications from the client 110 may be redirected via the network 115 from one component to another. The client 110 may run the message handling application 155 to handle the selection and presentation of the messages on the display via the user interface 160. The user interface 160 may be sized to fit the display form of the client 110. In some embodiments, when the client 110 is a mobile communication device (e.g., a smartphone or tablet), the user interface 160 may have a size corresponding to or matching size of the display of the client 110.

The message handling application 155 may be a web application to access functionalities provided by the mail management server 105. In some embodiments, the message handling application 155 may be an instance of an application for which the resources are hosted on the mail management server 105 of the cloud network 115. For instance, the message handling application 155 may be a part of a software as a service (SaaS) platform managed through the mail management server 105 and its components on the cloud network 115. With this arrangement, the functionalities of the message handling application 155 may be shared with the mail management server 105. The user interface 160 may have a message list (also referred herein as an electronic mailbox) to present a set of messages or threads received by the message handling application 155, a message panel to display the content of messages, and other user interface elements to manage the presentation of the messages, among others. The message handling application 155 may interface with the cloud computing network 115 to access the mail management server 105.

The mail management server 105 may be any computing device comprising one or more processors coupled with memory and software, and capable of performing the vari-

ous processes and tasks described herein. In some embodiments, the mail management server **105** may include a mail exchange server facilitating the delivery of electronic mail messages to the set of clients **110**. In some embodiments, the mail management server **105** may include one or more servers distributed in the cloud network **115**. The servers may be dedicated for interfacing with instances of the mail handling applications **155** on the clients **110**. For instance, the servers on the cloud network **115** may host and provide resources for web applications corresponding to the mail handling applications **155** accessed via the client **110**. The mail management server **105** may be in communication with the client **110**, via the network **115**. Although shown as a single mail management server **105**, the mail management server **105** may include any number of computing devices. The mail management server **105** may host, maintain, or otherwise carry various functions for mail exchange among the client **110**.

Within the mail management server **105**, the distribution list manager **120** may be or may include computer-readable machine code executable by the mail management server **105** to assign users to a distribution list composed of a subset of users. The assignment handler **125** may be, or may include, computer-readable machine code executable by the mail management server **105** to receive electronic messages and transmit them to a subset of users within a distribution list based on certain criteria. The message converter **130** may be, or may include, computer-readable machine code executable by the mail management server **105** to visually convert electronic mail conversation threads into a chat interface by removing auxiliary content from the electronic mail. The message converter **130** may also be, or include, computer-readable machine code executable by the mail management server **105** to integrate various chat platforms into a user's electronic mailbox as an electronic mail message. The attachment manager **135** may be, or may include, computer-readable machine code executable by the mail management server **105** to compile attachments within an electronic mail conversation thread and present the attachments to a user in a consolidated location.

In addition, the calendar manager **140** may be, or may include, computer-readable machine code executable by the mail management server **105** to determine a priority of calendar events and subsequently accept pending or declined events upon the occurrence of a predefined event. The response generator **145** may be, or may include, computer-readable machine code executable by the mail management server **105** to receive electronic mail messages, determine that the received electronic mail message contains a request, interpret the request, and automatically generate a responsive reply with relevant information. The metric tracker **150** may be, or may include, computer-readable machine code executable by the mail management server **105** to monitor at least one message center, track various predefined metrics based on characteristics of the electronic mail messages received by the at least one message center, and display the various tracked messages on user interface **160**. The message handling application **155** on the client **110** may share similar functionalities as the mail management server **105**.

The message handling application **155** executing on the client **110** may receive, retrieve, or otherwise identify one or more properties defining at least one distribution list to which to assign a user of the client **110**. The properties may identify or include, for example: a user identifier referencing the user, the client **110**, or the individual instance of the message handling application **155**; a request type of electronic mail messages the user is to be assigned to respond to;

a geographic region or location in which the user is located; group identifier referencing which group the user belongs to in an organization or association; and other information. The properties may be used to select which distribution list the user of the client **110** is to be assigned. At least a portion of the properties may be inputted by the user via the user interface **160** of the message handling application **155**. For example, upon request, the message handling application **155** may provide a dialogue box in the user interface **160** in which the user can input the properties for defining the distribution list. The dialogue box may be in the form as shown in FIG. 6.

With identification, the message handling application **155** may provide, send, or otherwise transmit the one or more properties to assign the user to the distribution list to the mail management server **105**. With detection of an interaction with the user interface **160** to input the properties for the distribution list, the message handling application **155** may send the one or more properties to the mail management server **105**. In some embodiments, the message handling application **155** may add information to the properties, outside of those inputted via the user interface **160**, upon detection of the interaction. For instance, upon completion of the dialogue box for defining the distribution list on the user interface **160**, the message handling application **155** may add the user identifier to the properties and transmit the properties to the mail management server **105**.

The distribution list manager **120** executing on the mail management server **105** may obtain, identify, or otherwise receive the one or more properties to assign the user of the client **110**. The one or more properties may be received via the user interface **160** of the message handling application **155** running on the client **110**. The distribution list manager **120** may process or parse the one or more properties to identify information (e.g., the request type or group identifier) inputted by the user of the client **110** and included by the message handling application **155**. Upon receipt, the distribution list manager **120** may store and maintain the properties on a database accessible to the mail management server **105** (e.g., on the cloud network **115**). In some embodiments, the distribution list manager **120** may automatically generate the parameters, independently from inputting via the mail handling application **155**. For example, the distribution list manager **120** may use organizational data identifying the users by associated group to produce the parameters defining which distribution list the user is to be assigned.

Using the one or more properties, the distribution list manager **120** may select or identify at least one distribution list to which to assign the user of the client **110**. The distribution list manager **120** may maintain a set of distribution lists for users accessing respective instances of the message handling application **155** running on clients **110**. Each distribution list may define or specify a combination of parameters (e.g., request types and group identifiers) with which to assign the user to the respective distribution list. To identify, the distribution list manager **120** may compare the one or more properties from the user with the specification of parameters for each distribution list. If the properties from the user do not match the specification, the distribution list manager **120** may refrain from assigning the user to the distribution list. Otherwise, if the properties from the user match the specification, the distribution list manager **120** may assign, set, or otherwise associate the user to the distribution list. The user may be assigned to one distribution list or multiple distribution lists.

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The assignment handler **125** executing on the mail management server **105** may identify or receive a set of electronic mail messages to be sent to the user of the client **110**. For example, the assignment handler **125** executing on the mail management server **105** operating as a mail exchange service can identify the electronic mail messages identifying the user of the client **110** as the recipient of the message. In some embodiments, the assignment handler **125** may intercept electronic mail messages in route to the client **110**. In some embodiments, the assignment handler **125** may retrieve, obtain, or identify the set of electronic mail messages already received at the client **110**. For instance, the assignment handler **125** may interface with a mail agent running on client **110** to identify the received electronic mail messages.

Each electronic mail message (sometimes herein referred to as an electronic mail or e-mail) may include at least one header and at least one body message. The header and body message may be in accordance with a syntax. The header may include, for example: a sender (e.g., a from field) identify the writer of the message; a timestamp (e.g., date, hour, minute, and seconds); one or more recipients (e.g., a To field, carbon copy (CC) field, and blind carbon copy (BCC) field) to which the message is to be sent; a message identifier uniquely identifying a particular electronic mail message or thread; a reply-to identifier corresponding to the message identifier corresponding to an electronic mail message which a subsequent electronic mail message is responding to; a subject including a descriptor of the body message; and content type identifying a formatting of the message (e.g., plaintext, rich text, or HyperText Markup Language (HTML), among others. The body message may include content corresponding to a message composed by the sender and to be viewed by the recipients.

Based on the assignment of the user to one of the distribution lists and the content in the electronic mail messages, the assignment handler **125** may select a subset of electronic mail messages for the user. From among the electronic mail messages to be sent to the user, the selected messages may correspond to those that are to be brought to the attention of the user. Each distribution may define or specify criteria for selecting the subset of electronic mail messages for the user. The criteria may include, for example, keywords or phrases correlated with a request type and group associated with the distribution list. For instance, for a financial institution, the distribution list associated with email addresses of securities brokers may include keywords or phrases, such as "purchase order," "order status," and ticker names, among others. Electronic mail messages with such keywords in the subject or body may be selected to bring to attention to the user of the mail management application **155**.

With the identification of the messages to be sent to the user, the assignment handler **125** may process or parse each electronic mail message to extract or identify the contents of the electronic mail message, such as the headers and the body message. Upon extraction, the assignment handler **125** may compare keywords from the content with the selection criteria for the distribution. Continuing from the previous example, the assignment handler **125** may scan the subject and body message to determine whether there are any keywords or phrases for the distribution list for securities brokers. If there are no content in the electronic mail message that match the selection criterion, the assignment handler **125** may exclude the electronic mail message from selection for the user. The assignment handler **125** may also determine that the electronic mail message is to be excluded

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from the distribution list to which the user is assigned. On the other hand, if there are content in the electronic mail message that does match the selection criterion, the assignment handler **125** may include or select the electronic mail message for the user. The assignment handler **125** may also determine that the electronic mail message is to be selected for the distribution list to which the user is assigned.

In some embodiments, the assignment handler **125** may apply a natural language processing (NPL) algorithm to the content of the electronic mail message. The NPL algorithm may include, for example, automated summarization, information retrieval, topic recognition, named entity recognition, or lexical semantics analysis (e.g., using a knowledge graph), among others to analyze the content of the electronic mail message. With the application of the NPL algorithm, the assignment handler **125** may produce or generate additional keywords, phrases, or other output from the content, with which to compare against the selection criterion of the distribution list. For instance, from applying the NPL algorithm to the body message, the assignment handler **125** may generate an output indicating that an electronic mail message contains content pertinent to security purchase orders. The assignment handler **125** may compare the output with the selection criterion of the distribution list as discussed above.

With the selection, the assignment handler **125** may provide, send, or otherwise transmit an identification of the subset of the electronic mail messages to the client **110** for presentation in an electronic mailbox of a user interface **160** of the message handling application **155**. The identification may reference the selected electronic mail messages to be brought to the user's attention, and may include instructions to the message handling application **155** to display the selected electronic mail messages separately from the unselected messages in the user interface **160**. In some embodiments, the assignment handler **125** may include the identification into the electronic mail message itself, for example, into the subject line or in the body message. In some embodiments, the assignment handler **125** may transmit the identification to the message handling application **155**, separately from the electronic mail messages.

From the mail management server **105**, the message handling application **155** may receive the identification of the subset of the electronic mail messages selected based on the assignment of the user to the distribution list and the content of the electronic mail messages. The message handling application **155** may receive the identification together with or separately from the electronic mail messages.

The message handling application **155** may display, render, or otherwise present the selected subset of electronic mail messages in a message list of the user interface **160**. The message list may be one or more elements of the user interface **160** for displaying the selected electronic mail messages as identified using the distribution list. The message list may catalogue the selected electronic mail messages (or threads) in a chronological sequence in order of sent or received time. The user interface **160** may also include message panel to present the content of one of the electronic mail message selected on the message list. The message list and message panel of the user interface **160** may, for example, be of the form in FIGS. 7-10. In some embodiments, the message list may include at least one element to enable the user to view all the electronic mail messages received by the user, including the messages excluded using the distribution list.

In some embodiments, the message handling application **155** may identify, from the subset of electronic mail messages, an electronic mail message for which no response is

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sent within a defined time window after receipt. Upon receipt of each electronic mail message, the message handling application **155** may maintain a timer to keep track of time of elapsed since receipt of the electronic mail message. The timer may be maintained for each electronic mail message selected using the distribution list. The defined time window may identify a span of time in which the message is to be responded by the user, for example, by creating and sending a response message. The time window may range, for example, between 3 days to 1 month. The message handling application **155** may compare the elapsed time to the defined time window. If the elapsed time does not exceed the time window, the message handling application **155** may continue to monitor for responses by the user to electronic mail message. If a response is detected within the time window, the message handling application **155** may deactivate or remove the timer for the electronic mail message.

If no response is detected within the time window, the message handling application **155** may initiate an escalation procedure. As part of the procedure, upon identification of the electronic mail message receiving no response within a defined time window after receipt, the message handling application **155** may present provide in the message panel of the user interface **160** an indicator. The indicator may identify the electronic mail message as having no response within the defined time window. The message handling application **155** may also send an indication to the mail management server **105** that another user in the same distribution list as the user is to be notified. Upon receipt, the assignment handler **125** on the mail management server **105** may identify one or more users in the same distribution list as the user. With the identification, the assignment handler **125** may provide the notification along with the electronic mail message itself to the other user.

In some embodiments, the assignment handler **125** on the mail management server **105** may identify the defined time window has passed with no response. The functionality of the assignment handler **125** may be similar to the message handling application **155** with respect to escalation as discussed above. The assignment handler **125** may identify, from the subset of electronic mail messages, an electronic mail message for no response is sent within a defined time window after receipt. Upon identification of an electronic mail message receiving no response within a defined time window after receipt, the assignment handler **125** may send or transmit an indicator to the client **110**. The indicator may identify the electronic mail message as having no response within the defined time window. Upon receipt, the message handling application **155** may present the in the message list or panel of the user interface **160**.

In some embodiments, the message handling application **155** may identify or receive from a user, via the user interface **160**, a request to change assignment of at least one electronic mail message of the subset of electronic mail messages to another user. The request may identify the electronic mail message for which the assignment is to be changed to another user or another distribution list. The request may be generated via a dialogue box of the user interface **160**, with which the user can identify the electronic mail message as well as one or more parameters (e.g., request type or group) for defining a distribution list or the specific distribution list. For instance, the user may use the user interface **160** to request that one or more electronic mail messages in the message list is to be reassigned another user.

Upon receipt, the message handling application **155** may provide, send, or otherwise transmit an indication that the electronic mail message is to be assigned to a second

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distribution list. The indication may be transmitted to the mail management server **105** in response to receipt of the request to change assignment. In conjunction, the message handling application **155** may remove the electronic mail message among the subset of electronic mail messages in the message list from presentation on the user interface **160**. The removed electronic mail message may be accessible through the user interface **160** when the user requests for display of all received electronic mail messages.

In some embodiments, the assignment handler **125** executing on the mail management server **105** may receive from a user, via client **110**, a request to change assignment of an electronic mail message of a subset of electronic mail messages to another user. The request may include or identify the electronic mail message as well as one or more parameters for defining a distribution list or the specific distribution list. In some embodiments, the assignment handler **125** may invoke the distribution list manger **120** to identify at least one other distribution list using the parameters from the request. In some embodiments, from invoking the distribution list manger **120**, the assignment handler **125** may identify at least one other user in the other distribution list to which to reassign the electronic mail message.

Upon identification, the assignment handler **125** may provide, send, or transmit an indication that the electronic mail message is to be assigned to the other user (or the other distribution list). The assignment handler **125** may then send an indication to client **110** to remove the electronic mail message among the subset of electronic mail messages in the message list from presentation in the user's electronic mailbox as displayed on user interface **160**. In addition, the assignment handler **125** may also send the indication to another client **110** of the identified user to which the electronic mail message is to be reassigned. The assignment handler **125** may provide, send, or otherwise forward the electronic mail message that is assigned to the client **110** of the other identified user.

With the presentation of the electronic mail messages, the message handling application **155** may detect, identify, or otherwise receive a request to present an electronic mail thread as a chat conversation thread, via the user interface **160**. The request may be detected via an interaction with a user interface element of the user interface **160**. The request may identify a set of electronic mail messages corresponding to an electronic mail thread that is to be converted to a chat conversation thread. The electronic mail thread may, for example, correspond to a set or chain of electronic mail messages among senders and recipients, with the same message identifier in the reply-to field.

Upon receipt of the request, the message handling application **155** may extract or identify main content and auxiliary content from the set of electronic mail messages corresponding to the electronic mail thread. Both the main content and the auxiliary content may be part of the body message of the set of electronic mail message. The main content may correspond to a portion of the body message including primary text and other content, such as the salutation and one or more subsequent paragraphs, generally toward the beginning and middle of the body message. The auxiliary content may correspond to the remaining portions of the message may, such as a closing message (e.g., "Best [name]"), signatures, and disclaimers generally toward the end of the body message.

In some embodiments, the message handling application **155** may recognize, detect, or otherwise identify the auxiliary content in the body message of each electronic mail message of the thread. For example, the message handling

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application **155** may use template defining text correlated with auxiliary content. The message handling application **155** may perform matching to compare against the contents of the body message. If there is a match, the message handling application **155** may identify the matching portions as auxiliary content, and the remaining portion as main content.

The message handling application **155** may generate a chat conversation thread to include the main content, while excluding the auxiliary content from the electronic mail messages corresponding to the electronic mail thread. In some embodiments, the message handling application **155** may add time stamps from the headers of the electronic mail messages forming the thread. The message handling application **155** may format the main content of the electronic mail thread to convert to the chat conversation thread. For instance, the message handling application **155** may set the font of the text in the main content in unison, remove indentations, and include time stamps, among other formatting changes to conform to specifications of the chat conversation thread. With the conversion, the message handling application **155** may present at least a portion of the chat conversation thread in the message panel of the user interface **160**. An example of the conversion from the electronic mail message to the chat conversation message is shown on FIGS. **16A** and **16B**.

In some embodiments, the message converter **130** executing on the mail management server **105** may perform the conversion of the electronic mail thread to present as the chat conversation thread. The conversion of the electronic mail thread to the chat conversation thread may be similar as discussed above. The message converter **130** may receive the request to present the electronic mail thread as the chat conversation thread via the interaction with the user interface **160** from the client **110**. Upon receipt of the request, the message converter **130** may identify main content and auxiliary content from a set of electronic mail messages corresponding to the electronic mail thread. The message converter **130** may generate a chat conversation thread to include the main content, while excluding the auxiliary content from the electronic mail messages corresponding to the electronic mail thread. The message converter **130** may send the chat conversation thread to the client **110** for presentation of at least a portion of the chat conversation thread in the message panel of the user interface **160**.

Conversely, the message handling application **155** may identify or receive a request to add a chat conversation thread from another application as one of the subset of electronic mail messages in the message list. The chat conversation thread may not be handled by the message handling application **155**, but by another application. The other application may be a chat conversation application, such as Symphony Communication™, Slack™, and WhatsApp™, among others. The message handling application **155** may interface with the chat conversation application to fetch, retrieve, or otherwise identify the chat conversation thread therefrom. The request may identify the particular chat conversation thread to be converted to a subset of electronic mail messages to form an electronic mail thread.

Upon receipt of the request, the message handling application **155** may output, produce, or otherwise generate an electronic mail thread using a set of conversations of the chat conversation thread. The conversation thread may identify participants. Each conversation may correspond to at least one message communicated from a sender to a recipient. To generate, the message handling application **155** may format the conversations of the chat conversation thread in accor-

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dance with the electronic mail thread. For instance, the message handling application **155** may use the identifiers for the sender and recipient fields of the header of a respective electronic mail message. In addition, the message handling application **155** may add the message of each chat conversation into the body message of the corresponding electronic mail message. With the generation, the message handling application **155** may display or present on the message list of the user interface **160** an identification of the electronic mail thread among a plurality of threads. Upon selection of the electronic mail thread in the message list, the message handling application **155** may display the corresponding set of electronic mail messages in the message panel of the user interface **160**. An example of the conversion from the chat conversation message to the electronic mail message is shown on FIG. **15**.

In some embodiments, the message converter **135** executing on the mail management server **105** may receive a request to add a chat conversation thread from a second application as one of the subset of messages in the message list. The addition of the chat conversation thread may be similar to the addition by the message handling application **155**, as discussed above. Upon receipt of the request, the message converter **135** may generate an electronic mail thread to include a second subset of messages corresponding to a plurality of conversations of the chat conversation thread. The message converter **135** may then provide the electronic mail thread to the mail management application **155** on the client **110** to present an identification of the electronic mail thread among a plurality of threads on the user interface **160**.

The message handling application **155** may identify a set of attachments (e.g., files or other messages) available to be added to the electronic mail messages in the message list of the user interface **160**. The message handling application **155** may provide one or more user interface elements to select the set of attachments via the user interface **160**. For instance, the message handling application **155** may provide a file directory window via which the user can navigate the file hierarchy of the client **110** or other cloud storage services to select files for attachments. Upon selection, the message handling application **155** may provide the set of attachments available to be added to at least one of the electronic mail messages via an attachment window at least partially concurrently with the message list of the user interface **160**.

In some embodiments, the attachment manager **135** executing on the mail management server **105** may provide, via an attachment window at least partially concurrently with the message list of the user interface, a set of attachments available to be added to at least one of the subset of messages. The provision of the set of attachments may be similar as discussed above. The attachment manager **135** may receive the selection of attachments via the user interface **160** of message handling application **155**. Upon selection, the message handling application **155** may provide the identification of the attachments for presentation via a separate window or list on the user interface **160**.

The message handling application **155** may identify or receive a request to cancel a first calendar event via the user interface **160**. The calendar event may correspond to one of the subset of electronic mail messages that was previously received and accepted by the user of the client **110**. The request itself may identify the calendar event to be canceled. For instance, the user of the message handling application **155** may interact with a user interface element corresponding to the calendar event to open up a window identifying

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details of the calendar event to cancel the calendar event. The message handling application **155** may detect the interaction with the window to cancel as the request to cancel.

Upon receipt of the request, the message handling application **155** may identify a second calendar event for a time slot at least partially concurrent with the first calendar event. The message handling application **155** may identify selected electronic mail messages corresponding to calendar events. From the identified calendar events, the message handling application **155** may identify another calendar event at least partially overlapping with the canceled calendar event. With the identification of the calendar event, the message handling application **155** may present an identification of the calendar event as a recommendation to replace the canceled calendar event on the user interface **160**.

In some embodiments, the calendar manager **140** executing on the mail management server **105** may receive a request to cancel a first calendar event corresponding to one of the subset of electronic mail messages. The handling of the request to cancel by the calendar manager **140** on the mail management server **105** may be similar to the handling of the request to cancel as discussed above. Upon receipt of the request, the calendar manager **140** may identify a second calendar event for a time slot at least partially concurrent with the first calendar event. With the identification of a second calendar event, the calendar manager **140** may provide on the user interface **160** the identification of the second calendar event.

The message handling application **155** may provide a response for at least one of the electronic mail messages in the user interface **160**. The response may be automatically generated using at a portion of the content (e.g., the header or the body message) of the electronic mail message. In some embodiments, the message handling application **155** may detect, identify, or otherwise receive a request to provide an automated response via the user interface **160** or from the content of the electronic mail message itself. The request may identify the electronic mail message for which the automated response is to be generated. The response may be provided as a draft electronic mail message for the user of the client **110** to reply to the sender.

Using the content of the electronic mail message, the message handling application **155** may generate a query for information to send to a search platform (e.g., for searching an internal dataset). The query may be generated in response to presence of keywords or phrases within the content correlated with the query for information. For instance, the message handling application **155** may identify that the electronic mail message includes keywords associated with a particular order of securities, and may generate query for the platform using the keywords and identification of the order. The message handling application **155** may receive results from the platform, and use the results to generate the automated response. Examples of the automated generated response may be of the form as seen in FIGS. **17** and **18**.

In some embodiments, the response generator **145** executing on the mail management server **105** may generate and provide an automated response for at least one of the electronic mail messages in the user interface **160**. The generation of the automated response may be in a similar manner as discussed above. The response may be automatically generated using at a portion of the content (e.g., the header or the body message) of the electronic mail message. In some embodiments, the response generator **145** may detect, identify, or otherwise receive a request to provide an automated response via the user interface **160** or from the contents of the electronic mail message. Using the content of

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the electronic mail message, the response generator **145** may generate a query for information to retrieve from a search platform. With the results from the platform, the response generator **145** may generate the automated response.

The message handling application **155** may provide a statistic indicator associated with provision of the electronic mail messages to the client **110** in accordance with the distribution list. The statistic indicator may correspond to visualization of the statistics and metrics associated with the electronic mail messages for the client **110**, the user, or the overall distribution list, such as percentages of open inquiries, request types, escalations, response rate, and trends, among others. The statistic indicators may be provided upon request by the user, for example, by interaction with the user interface **160** to retrieve statistics. The message handling application **155** may present the statistic indicators in a dashboard of the user interface **160**. In some embodiments, the metric tracker **150** executing on the mail management server **105** may provide and present statistics in a similar manner as discussed. Examples statistics are shown on FIGS. **3-5**.

FIG. **2** depicts a flow diagram of a process **200** of managing and handling electronic mail messages. Embodiments may include additional, fewer, or different operations from those described in the process **200**. The process **200** may be performed by any of the components described herein, for example, the mail management system **105** and the client **110** of FIG. **1**. At step **205**, a computing system may identify properties defining a distribution list for assigning a user. The properties may identify or include, for example: a user identifier referencing the user, the client, or the individual instance of an application; a request type of electronic mail messages the user is to be assigned to respond to; a geographic region or location in which the user is located; group identifier referencing which group the user belongs to in an organization or association; and other information.

At step **210**, the computing system may identify a distribution list for a user. The computing system may maintain a set of distribution lists to which to assign users. Each distribution list may define or specify a combination of parameters (e.g., request types and group identifiers) with which to assign the user to the respective distribution list. In identifying, the computing system may compare the one or more properties from the user with the specification of parameters for each distribution list. If the properties from the user do not match the specification, the computing system may refrain from assigning the user to the distribution list. Otherwise, if the properties from the user match the specification, the computing system may assign, set, or otherwise associate the user to the distribution list.

At step **215**, the computing system may select a subset of electronic mail messages for the user in accordance with the distribution list. The selection may be from electronic mail messages intended to be received by the user. Each electronic mail message may include a header (e.g., sender, recipients, message identifier, and subject) and a body message, among others. Each distribution list may specify criteria (e.g., keywords or phrases) for selecting electronic mail messages. The computing system may perform the selection by scanning the electronic mail message to determine whether the message contains any matching keywords or phrases. If there are matching keywords or phrases, the computing system may select the electronic mail message.

At step **220**, the computing system may present the selected electronic mail messages in a user interface. The computing system may provide an identification of the

selected electronic mail messages. The identification may reference the selected electronic mail messages to be brought to the user's attention in a message list. The computing system may also provide instructions to the application to display the selected electronic mail messages separately from the unselected messages in the user interface.

At step 225, the computing system may perform various operations on the electronic mail messages on the user interface. Upon presentation, the computing system may provide one or more functionalities with respect to the electronic mail messages. For instance, upon request, the computing system may provide an email to chat conversion, a chat to email conversion, automated response generation, calendar managements, and file attachments, among others.

Referring now to FIG. 3, depicts a screenshot of a dashboard graphical user interface 300 of mailbox statistics displayed via an interface. The GUI 300 may include various graphics to display certain message statistics 302-310 to a user via a computing device. As depicted, the message statistics displayed on GUI 300 may include open inquiries by group 302; inquiries trend 304; mailbox statistics 306, including pending approval 306a, potential escalation 306b, assigned to me 306c, and specific client 306d; open inquiries by request type 308; and intensity heatmap 310, among others.

The open inquiries by group 302 may include a graphical illustration 302a to depict each group's open inquiries as a percentage of all open inquiries. In some embodiments, the graphical illustration 302A is a pie chart. In other embodiments, the graphical illustration 302A is any graphical illustration that can be used to compare percentages of a whole (e.g., a bar graph, a histogram, a scatter plot, etc.). The Open Inquiries by Group may also include a legend 302b for the graphical illustration 302A. Legend 302B may use various visualizations to map the information based in the graphical illustration 302A to the legend information found in legend 302B.

The Inquiries Trend 304 may include a graphical illustration 304A to depict the number of messages received over time in a group. In some embodiments, the graphical illustration 304A is a timeline. According to some embodiments, the x-axis of the graphical illustration 304A is time and the y-axis is the number of messages received. Additionally, in some embodiments, a time option 304A to change the time period displayed on the x-axis. In some embodiments, a user is able to choose between various time periods (e.g., one day, one week, one month, one year, etc.) for which to view the number of messages received by selecting one of the options available at time option 304A.

Upon selecting a time period, graphical illustration 304A may be automatically rescaled to the appropriate time period, corresponding to the option chosen by user. In some embodiments, the x-axis of graphical illustration 304A may be divided into several smaller time periods to aid in viewing the data. The GUI 300 may be divided along the x-axis automatically based on an appropriate division of time with respect to the option chosen by the user at option 304B. In some embodiments, the user may select a custom time division for the x-axis. In addition, the inquiries trend 304 may also include an inquiry option 304C. Inquiry option 304c may include a visual element with which the user may interact with to select a specific message or inquiry type to display statistics on. The statistics displayed may be the timeline of number of messages/inquiries received over time with the chosen message or inquiry type.

The mailbox statistics 306 may include a group's or individual's various mailbox statistics 306A-306D, including Pending Approval 306A, Potential Escalation 306B, Assigned to Me 306C, and Specific Client 306D. Mailbox statistics 306A-D may include a graphical illustration to aid in displaying the statistics. In addition, the mailbox statistics 306A-D may include a total number of messages relating to the corresponding statistic with a change in percentage from a previous time period. In some embodiments, the change in percentage may be color-coded so as to easily communicate to the user whether a particular statistic increased or decreased. For example, a positive percentage change (e.g., qualitative or quantitative) may be displayed in a first color (e.g., green font). Conversely, a negative percentage change (e.g., qualitative or quantitative) may be displayed in a second color (e.g., red font). Also, a static percentage (e.g., a percentage change of "0") may be displayed in a third color (e.g., yellow font).

The open inquiries by request type 308 may include a graphical illustration 308A. The graphical illustration 308A may display the number of messages/inquiries received by an individual or group, divided by the type of message or inquiry. For example, graphical illustration 308A may be a bar graph with various distinct message types and displays the number of each message type by quantity received, as shown in FIG. 3. According to other embodiments, the graphical illustration 308A may be an alternative graph type (e.g., pie graph, scatter plot, histogram, etc.). As with open inquiries by group 302, open inquiries by request type 308 may include a legend 308B to aid the user in deciphering the data displayed in graphical illustration 308A. Open inquiries by request type 308 may also include an option 308C to hide the legend 308B. Option 308C may be visual element capable of receiving various inputs. For example, option 308C may be a radio button, text string input, slider, dial, or drop-down menu, among others.

The Intensity Heatmap 310 may also include various graphical illustrations 310A-C. The Intensity Heatmap 310 may depict the amount of messages having a particular priority handling, such as a highest priority 310a (e.g., platinum), a second priority 310b (e.g., priority), and a third priority or other status 310c. The Intensity Heatmap 310 may be useful in illustrating how the messages are allocated to the various priority statuses, which may be helpful in allocating the workflow.

Referring now to FIGS. 4A and 4B, each depict a screenshot of a dashboard of mailbox statistics displayed via an interface for a mobile device. The statistical information displayed the mobile device. The webserver or local hosting agent may display the statistical dashboard 400A-B through a web-based browser on the user's mobile device. The statistical data (with accompanying options and legends) displayed on statistical dashboard GUI 300 of FIG. 3 may be displayed on statistical dashboards 400A-B. As depicted here, the statistical dashboard 400A-B may be sized to fit the form of a mobile device. For example, the mailbox statistics 402-406 may be vertically displayed so as to allow the user to scroll vertically on a mobile device to view the mailbox statistics 402-406.

Referring now to FIG. 5, depicted is a screenshot of a dashboard GUI 500 of various statistics displayed via an interface for a mobile device. The statistical dashboard GUI 500 may be displayed in a web-based browser on a computing device. The statistical dashboard GUI 500 may include several statistical analyses 502-532 of a given mailbox. The mailbox may represent a single user's mailbox, a distribution mailbox, a group mailbox, etc. As such, the

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statistical analyses **502-532** may communicate information on both individual mailboxes and distribution/group mailboxes. Statistical analyses **502-532** may display statistics on (1) messages unassigned for greater than one hour **502**, (2) messages responded to in less than two hours **504**, (3) messages resolved in less than twenty-four hours **506**, (4) total external messages **508**, (5) client potential escalations **510**, (6) messages in the queue for more than two hours **512**, (7) client inquiries open for more than twenty-four hours **514**, (8) snoozed inquiries **516**, (9) total activity **518**, (10) messages aged over three days **520**, (11) total users **522**, (12) closed inquiries **524**, (13) a comparison between users responding to messages in less than two hours **526**, (14) a comparison between users resolving to messages in less than twenty-four hours **528**, (15) a comparison between users' total messages **530**, and (16) messages dead on delivery **532**, among others.

The detailed statistical dashboard GUI **500** may prioritize statistics based on metrics tracking a clients' experience rather than the user's productivity by displaying client-experience statistics first as shown in the statistical analyses **502-516**. Overall statistics **518-524** may be displayed under the client-experience statistics **502-516**. The user may also customize the view of the detailed statistical dashboard by choosing which statistical analyses **502-532** to include, in what order to include, and what format to display. As with the GUI **300** of FIG. 3, the statistical analyses **502-532** may use graphic illustrations to display the statistical information provided on the dashboard. For example, a percentage graph **502a** may use two colors to show what percentage of total messages meet a specified criteria (e.g., number of messages unassigned for more than two hours).

Likewise, the graphical illustration may include a change-in-percentage indicator **502b**. The change-in-percentage indicator **502b** may use a color-coded format to visually display changes in percentage. For example, change-in-percentage indicator **502b** may use a first font color (e.g., green) to signify a positive change in percentage of messages unassigned for over an hour (e.g., positive qualitatively or quantitatively). A second font color (e.g., red) may be used to signify a negative change in percentage of messages unassigned for over an hour (e.g., negative qualitatively or quantitatively). A third font color (e.g., yellow) may be used to signify no change in percentage of messages unassigned for over an hour (e.g., no change qualitatively or quantitatively).

Detailed statistical dashboard GUI **500** may also include statistical analyses **526-530**, wherein multiple users' statistics are comparatively displayed. For example, user-comparison graph **526A** may display, in a bar graph, a subset of users with corresponding data related to a specified organizational metric. For example, in user-comparison graph **526A**, users are compared based on how many messages each user responded to within two hours of receipt. In this graphical illustration, a user (e.g., a manager or administrator) may more readily track individual users' metrics when compared to other users. If the number of selected users to compare is too large to fit within the predefined space for comparison graph **526A**, the user may scroll within the predefined space for comparison graph **526A** to view all available data. A user may customize the detailed statistical dashboard GUI **500** by defining the set of users or groups to compare. The user may define the set of users or groups to compare in a variety of ways, including radio buttons, drop-down menus, text string input, etc. In this way, the user

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may view one subset of users in one viewing configuration, and then view a second subset of users in a second viewing configuration.

Referring now to FIG. 6, depicted is a screenshot of a dialogue box **600** for selecting parameters to define a distribution list to which to assign a user. The dialogue box **600** may be for specifying request types and ownership for a distribution list **602**. As illustrated, a user may define the criteria (i.e., "request type") **606** for a message to be assigned to a distribution list **602**. A user may also specify the owners **604** for a distribution list **602**.

In some embodiments, a user may select a distribution list **602** from a list of distribution lists found within the distribution list dialogue box **600**. For example, dropdown menu **610** may include all available distribution lists for a user to select. Once a distribution list (e.g., **602**) is selected, a user may define the members of the distribution list (i.e., owners **604**). Selected owners **604** receive all messages meeting distribution list's message characteristics **606** (i.e., request type).

In distribution list dialogue box **600**, the user may also define the distribution list's message characteristics **606** of the messages assigned to distribution list **602**. For example, a user may define a distribution list message characteristic **606** to be certain content within the body of a message (e.g., inquiries on a financial trade, bank statement, credit card inquiry, lending request, etc.). Distribution list message characteristics **606** may also include certain recipients (e.g., all messages going to a team lead may be sent to the team distribution list), certain senders (e.g., all messages coming from a client may be sent to the client distribution list), time or receipt (e.g., all messages received after hours may be sent to the after-hours distribution list), etc.

Referring now to FIG. 7, depicted is a screenshot of an interface **700** of an application including a message list and a message panel. The interface **700** may include an electronic mailbox **702** (also herein referred to as a message list) with accompanying message panel **704**. In this configuration, the messages selected using the Distribution List Manager **120** may be displayed in the user's electronic mailbox **702** vertically in a list. Because the messages are selectively sent to the user based on artificial intelligence, natural language processing, machine learning, and user input, a reduction of junk mail and irrelevant messages are received by the user.

The provision of electronic mail messages in this manner may drastically reduce the amount of mail the user receives. Because of this, the user may utilize the electronic mailbox **702** as a "to-do" list, with only actionable and relevant emails arriving at the inbox. Various relevant data categories are displayed for each email. For example, "from" **702A**, "subject" **702B**, "assigned to" **702C**, "assigned from" **702D**, "received" **702E**, "age of message" **702F**, "possible escalation" **702G**, among others are displayed.

The electronic mailbox **702** may be configurable to be filtered according to the various relevant data categories. Additionally, the user may click and drag individual messages within the list of messages to customize the order of messages to the user's preference. In this way, the user may further utilize the electronic mailbox **702** as a "to-do" list by deciding which messages reside at the top of the list of messages, and thus may command more attention of the user. In addition to the electronic mailbox **702**, the interface **700** includes a message panel **704**. Once a user selects a message **708** from the electronic mailbox **702**, the full message populates in the message panel **704**.

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A user's electronic mailbox may also be displayed by the mail management server **105** of FIG. **1** in a mobile application. As with the message center **700** of FIG. **7**, the mobile electronic mailbox of FIGS. **8-10** may be set to various predefined filtered views which may filter the user's messages from various messaging platforms (e.g., email, chat, text message, etc.) for specific work flows. For example, FIG. **8** depicts an electronic mailbox **800** in a filtered "Assigned to Me" view, FIG. **9** depicts an electronic mailbox **900** in a filtered "Pending Approval" view, and FIG. **10** depicts an electronic mailbox **1000** in a filtered "Potential Escalation" view. Additional filtered views exist, including user-customized filter views. In these filtered views, select messages meeting the filtered view's filtering criteria are shown. In this way, a user's electronic mailbox shows only the relevant messages to a user.

Referring now to FIG. **8**, depicted is a screenshot of an interface **800** of an application for a mobile device including a message list. As in the interface **700**, the interface **800** may display the messages selected using the distribution list. Various relevant data categories may be displayed for each message, for example "from," "subject," "assigned to," "assigned from," "received," "age of message," "possible escalation," "read/unread status," "attachments," "type of message" (e.g., conversation thread, email, chat, etc.), among others. The interface **800** is configurable to be filtered according to the various relevant data categories described above. Additionally, the user may click and drag individual messages within the list of messages to customize the order of messages to the user's preference. In this way, the user may further utilize the interface **800** as a "to-do" list by deciding which messages reside at the top of the list of messages, and thus may command more attention of the user. The interface **800** may include the filtered "Assigned to Me" view **802**, according to one embodiment. The messages specifically assigned to the user may be filtered for the user to view.

Referring now to FIG. **9**, depicted is a screenshot of an interface **900** of an application for a mobile device including a message list with indicators regarding approval. If there are no messages assigned to the user then the user is presented with a second filtered view; the "Pending Approval" view **902** of FIG. **9**. In some embodiments, a user may select for a sent message to remain pending until a second user approves the message for sending. A user may elect to request approval before sending a message for various reasons. For example, a user may elect to request approval when a message contains confidential information, a message is to an important client, the message has an attachment that is to be reviewed, etc.

In the "Pending Approval" view **902**, a user may view messages that are pending that user's approval. These messages, while resident in a first user's "Pending Approval" view **902**, may be in multiple user's "Pending Approval" view, per the use of distribution lists manager **120** and assignment handler **125** as described herein. In this way, various users may award approval to the pending approval requests. Once approval has been awarded, the message may be hidden from the mailbox of each user who was assigned the message. The interface **900** may include a characterization of the source of each message displayed. For example, email icon **904B** may characterize the message **904A** as being an email while chat message **906B** may characterize the message **906A** as being a chat message from a distinct messaging platform.

Referring now to FIG. **10**, depicted is a screenshot of an interface **1000** of an application for a mobile device includ-

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ing a message list with indicators for potential escalations. The interface **1000** may be in a third, filtered view, a "Potential Escalation" view **1002**. In some embodiments, various messages may be determined to be escalated in an electronic mailbox when certain criteria are met. For example, if an actionable message (i.e., a received message requesting a response) has not been responded to in a predefined set of time (e.g., within twenty-four hours), the message may be escalated.

In other words, the message may be sent to more users within a distribution list for which the message was previously hidden. These other users may be coworkers in the same distribution list, or even a manager. This may ensure that a message is responded to in due course by bringing attention to a pending message. To avoid this escalation, a precursor to escalation, called "potential escalation," may be carried out. In some embodiments, a message may be sent to (or unhidden from) additional users within a distribution list if an actionable message is not responded to within a predefined set of criteria. For example, if a message will be escalated at twenty-four hours after receipt, it may become potentially escalated at two hours to ensure an escalation does not occur.

In the "Potential Escalation" view **1002** (as well as in the other filtered views), if a message is at risk of potential escalation, the escalation may be indicated by a visual element **1004**. In some configurations, the visual element is a red triangle with an exclamation point within. The visual element **1004** can be any indicator to draw the user's attention to the potential escalation risk, including animations, graphics, sounds, etc. The visual element may be an interactive element, so that when a user hovers an input device (e.g., a mouse cursor) over the visual element **1004**, the reason for the risk of potential escalation appears in a dialogue box (e.g., too much time has passed since receipt of the message.). In "Potential Escalation" view **1002**, a user may be presented only with messages are characterized.

In some embodiments, the users will be directed to the "Potential Escalation" view **1002** if there are no messages in the "Assigned to Me" view **802** of FIG. **8** or the "Pending Approval" view **902** of FIG. **9**. In this way, a user may be directed to the most relevant messages in a predefined sequential order. The criteria for the various filtered views described above may be defined by the user or the organization (e.g., through a manager, administrator, etc.). Likewise, the default view (and subsequent order of displayed views) may also be defined by the user or the organization (e.g., through a manager, administrator, etc.). Likewise, a user may define their own personalized filtered view based on the various message data described above (e.g., to, from, subject, etc.).

Referring now to FIG. **11**, depicted is a screenshot of an interface **1100** of an application for a mobile device including a message with various tags. By using various input fields **1102A-F**, a user may tag a message within their electronic mailbox. The tagging may occur through any variation of input selection including drop-down menus, text string input, radio buttons, etc. From tagging a message, a user may manually move messages between distribution lists, update message characteristics, etc. For example, upon receipt of a message, the mail management server **105** may assign the message to User 1 based on (1) the contents of the message and (2) the roles of users within Distribution List 1 (of which User 1 is a member). That message would appear in User 1's electronic mailbox, specifically in User 1's "Assigned to Me" view **802** of FIG. **8**. User 1 may realize that the message is not meant for the User 1, but rather, for

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a different group of users in Distribution List 2. User 1 may open the message's tagging dialogue box **1100** to change the Group Tag **1102E**. Upon changing the Group Tag **1102E**, the message will be unassigned from User 1 (and the rest of the assigned users in Distribution List 1) and reassigned to a subset of users within Distribution List 2 who are most likely to be responsible for the message and any corresponding tasks.

Once the mail management server **105** assigns an actionable message (i.e., an email requiring a response) to a user, the user (or other assigned user) may be assigned to resolve the assignment by completing the assigned task or otherwise dispatching it (e.g., assigning it to another user, tagging the message as not actionable, etc.). In some embodiments, actionable messages are called "queries." In some embodiments the mail management server **105** automatically may mark an actionable message as resolved based on a user's interaction with the message (e.g., if the user assigns the message to another user or distribution list, if the user responds to the email, if the user deletes the email, etc.).

Referring now to FIG. **12**, depicted is a screenshot of a dialogue box for an inquiry resolution. A user may manually tag a message as resolved once the user has dispatched of the actionable message (i.e., the query). The Resolve Inquiry dialogue box **1200** is shown. A user may arrive at the dialogue box **1200** from the user's electronic mailbox (as shown in FIGS. **8-10**) and selecting a resolve-inquiry visual element embedded in the page that corresponds to the actionable message. The user may then use the various input fields **1202-1208** to tag the actionable message as dispatched. If the user does not need to reply to an inquiry, the user can mark the inquiry as resolved, such as checkbox **1208** for "actioned—no reply." The action of resolving can also automatically generate an email message to interested parties that confirms the resolution of the inquiry. By resolving an inquiry, the inquiry is removed from being an actionable item that appears for each user.

Referring now to FIG. **13**, depicted is a screenshot of an interface **1300** of an application including a message list and a message panel with a side panel including an audit of events. In the interface **1300**, an audit timeline **1304** is displayed in conjunction with other user interface elements. In this configuration, the audit timeline **1304** may be displayed for viewing in the user's message center **1300**. The audit timeline **1304** may be a snapshot of all events that have occurred in the life of the message from receipt, and may include entries for every event occurring therein. For example, once a message is received by the mail management server **105**, a first entry may be created in the timeline. This event contains and displays relevant data concerning the message for the user to view in relation to the rest of the timeline. This data may include the sender, the recipient, the subject, time of receipt, whether the message is internal or external, etc. By selecting the message within the timeline, the user may access additional data not displayed in the default view. This additional information may include any attachments included with the message, the actual text of the message, any embedded metadata, geographical location of sender and recipient, etc. Once an action is performed with respect to the message (e.g., responding, adding a note, forwarding, etc.) a new event **1306** is created in the timeline with all corresponding information.

A user may create and access a file containing all of the information found in the audit timeline **1304**. In some configurations, there is a first visual element **1308A**, through which a user may download a spreadsheet of the audit-timeline information by selecting the first visual element

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1308A with a user's input device (e.g., a mouse cursor). In some configurations, there is also a second visual element **1308B**, through which a user may download a PDF file containing the audit-timeline information by selecting the second visual element **1308B** with a user's input device (e.g., a mouse cursor).

FIG. **13** depicts an Auto Assign button **1310**. When a message is delivered to a distribution list having a plurality of users, the message may be automatically assigned to a particular user or group of users. The message will be automatically moved from the inbox to an Auto Assign folder. Removing the message from the inbox allows users to more efficiently handle those messages that have not been assigned to others and may still need resolution. Adding the message to the Auto Assign folder allows the users to check the folder to confirm that the message has been assigned to that user or another user. As a result, the Auto Assign functionality may further increase efficient use of this mailbox system. The message may appear in the inbox, but the system will automatically remove the message and present it in a different box for viewing so that the user does not need to respond to the message in the inbox.

Referring to FIGS. **12** and **13**, in one embodiment, an artificial intelligence model may be trained on previous inquiries and resolutions. The artificial intelligence model may automate an assignment of an inquiry based on how previous inquiries were routed and handled until resolution. The artificial intelligence model may also be trained to respond with a resolved confirmation or to mark as resolved without a reply. The artificial intelligence model may be configured to automatically assign inquiries and automatically move messages from an inbox to an Auto Assign box as appropriate.

Referring now to FIG. **14**, depicted is a screenshot of an interface **1400** of an application including a message list and a message panel with a side panel including notes. In the interface **1400**, a note-taking feature **1404** is displayed in conjunction with other user interface elements. In this configuration, the note-taking feature **1404** is displayed for viewing in the user's message center **1400**. The note-taking feature **1404** is available for the user to include notes in conjunction with a selected message **1402**. Any notes created by a user in conjunction with a selected message **1402** are attached to the selected message **1402** for internal users to view. Notes added through the note-taking feature **1404** may be accessible by internal users of the mail management server **105**. In this way, internal users may communicate with each other about a selected message without creating additional messages within the user's message center **1400** or exposing their comments to external parties.

Referring now to FIG. **15**, depicted is a screenshot of an interface **1500** of an application including a message list and a message panel with an electronic mail thread converted to a chat conversation thread. In this configuration, a user may integrate various chat platforms **1504** with the mail management server **105**. With this integration, a user may monitor various chat messaging platforms in the same location as the user's electronic mail. The user may filter the electronic mailbox **1502** to show only chat messages by selecting a chat visual element **1506**. In other embodiments, the electronic mailbox **1502** may show electronic mail and chat messages in a single inbox. Using the chat integration **1508**, the user may replay to chat messages from a separate chat platform while from inside the message center **1500**. This allows the user to monitor and reply to electronic messages from any platform in a single location, thus increasing efficiency and productivity for the user.

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Referring now to FIG. 16A, depicted is a screenshot of a selected electronic mail thread **1600A** with multiple messages. The thread **1600A** may be between a first participant **1602** and a second participant **1604**. In this configuration, the email thread **1600** may contain various conversationally extraneous elements (i.e., auxiliary content) **1606A-1610A** in addition to the conversationally relevant elements (main content) **1612A-1616A**. With this example, the main content **1612A-1616A** may be between within the auxiliary content **1606A-1610A**.

Referring now to FIG. 16B, depicted is a screenshot of a chat conversation thread **1600B** converted from the electronic mail thread. The chat conversation thread **1600B** between a first participant **1602B** and a second participant **1604B**. In the chat conversation thread **1600B**, the auxiliary content **1606A-1610A** of an email may have been removed from view so as to display the email chain as a chat message. In this configuration, the user is able to clearly view the main content **1612B-1616B** of the email chain without the extraneous distractions. In this way, the user's productivity may be increased by easing the strain of parsing through auxiliary content, or conversational "refuse." In addition, in some configurations the chat conversation thread **1600B** displays the messages as a chat with the most recent message **1620B** at the bottom of the thread, as displayed. In the chat conversation thread **1600B**, the user may still be using electronic mail (as opposed to a chat platform). The user may reply to the client with email by inputting text in the text string input **1622B** and selecting the "Send" graphical input **1624B**. While the text string input **1622B** and "Send" graphical input **1624B** appear as a chat platform, the user replies through electronic mail.

Referring now to FIG. 17, depicted is a screenshot of a dialogue box **1700** for suggested automated responses. In response to a client inquiry, an automated reply may be generated for the user to include in responding to the client. An artificial intelligence (AI) model can be trained to produce automated responses to a variety of client inquiries. In the example, an automated response message **1702** may be generated in response to a client's trade-status inquiry. The automated response message **1702** may be comprised of an introduction **1704**, colloquial pleasantries **1706**, and the inquiry response **1708**. The inquiry response **1708** may include answers several distinct inquiries within a single reply message. The dialogue box **1700** may also include response attachments **1712** that the user may elect to attach to the automated response message **1702**. By selectin, a visual selection element **1710**, the user may attach relevant data or attachments to the automated response message **1702**. Referring now to FIG. 18, depicted is screenshot of an automated reply included in a response electronic mail message **1800**. The response message **1800** may include the automated response message **1702** may be generated in response to a client's trade-status inquiry.

The foregoing method descriptions and the process flow diagrams are provided merely as illustrative examples and are not intended to require or imply that the steps of the various embodiments must be performed in the order presented. The steps in the foregoing embodiments may be performed in any order. Words such as "then," "next," etc. are not intended to limit the order of the steps; these words are simply used to guide the reader through the description of the methods. Although process flow diagrams may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process may correspond to a method, a function, a proce-

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cedure, a subroutine, a subprogram, and the like. When a process corresponds to a function, the process termination may correspond to a return of the function to a calling function or a main function.

The various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the embodiments disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

Embodiments implemented in computer software may be implemented in software, firmware, middleware, microcode, hardware description languages, or any combination thereof. A code segment or machine-executable instructions may represent a procedure, a function, a subprogram, a program, a routine, a subroutine, a module, a software package, a class, or any combination of instructions, data structures, or program statements. A code segment may be coupled to another code segment or a hardware circuit by passing and/or receiving information, data, arguments, parameters, or memory contents. Information, arguments, parameters, data, etc. may be passed, forwarded, or transmitted via any suitable means including memory sharing, message passing, token passing, network transmission, etc.

The actual software code or specialized control hardware used to implement these systems and methods is not limiting. Thus, the operation and behavior of the systems and methods were described without reference to the specific software code being understood that software and control hardware can be designed to implement the systems and methods based on the description herein.

When implemented in software, the functions may be stored as one or more instructions or code on a non-transitory computer-readable or processor-readable storage medium. The steps of a method or algorithm disclosed herein may be embodied in a processor-executable software module, which may reside on a computer-readable or processor-readable storage medium. A non-transitory computer-readable or processor-readable media includes both computer storage media and tangible storage media that facilitate transfer of a computer program from one place to another. A non-transitory processor-readable storage media may be any available media that may be accessed by a computer. By way of example, and not limitation, such non-transitory processor-readable media may comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other tangible storage medium that may be used to store desired program code in the form of instructions or data structures and that may be accessed by a computer or processor. Disk and disc, as used herein, include compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk, and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media. Additionally, the operations of a method or algorithm may reside as one or any combination or set of codes and/or instructions on a non-transitory

processor-readable medium and/or computer-readable medium, which may be incorporated into a computer program product.

The preceding description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the present disclosure. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the disclosure. Thus, the present disclosure is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the following claims and the principles and novel features disclosed herein.

While various aspects and embodiments have been disclosed, other aspects and embodiments are contemplated. The various aspects and embodiments disclosed are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

What is claimed is:

1. A method, comprising:

receiving, by a server, from a client device, a request to present an electronic mail thread corresponding to a subset of electronic mail messages in a plurality of electronic mail messages in a first message panel of a user interface in an application, as a chat conversation thread in a second message panel of the user interface; identifying, by the server, from each of the subset of electronic mail messages, main content and auxiliary content, responsive to the request to present; generating, by the server, the chat conversation thread including the main content identified from each of the subset of electronic mail messages, while excluding the auxiliary content; sending, by the server, to the client device, the chat conversation thread to be presented in the second message panel of the user interface in the application and to remove the subset of electronic mail messages from presentation in the first message panel; receiving, by the server, from the client device, an input on the second message panel corresponding to a reply message to at least one of the subset of electronic mail messages; and causing, by the server, the client device to present the reply message as part of the chat conversation thread on the second message panel.

2. The method of claim 1, further comprising

receiving, by the server, from the client device, a request to add a second chat conversation thread as one of the plurality of electronic mail messages in the first message panel; generating, by the server, an electronic mail thread to include a second subset of electronic mail messages corresponding to a plurality of conversations of the second chat conversation thread; and sending, by the server, to the client device, an identification of the electronic mail thread among a plurality of threads for presentation the message panel.

3. The method of claim 1, wherein generating the chat conversation thread further comprises generating the chat conversation thread in accordance with a format defining presentation of the chat conversation thread in the message panel.

4. The method of claim 1, further comprising identifying, by the server, from a second application separate from the

application, a second chat conversation thread to be converted to a second subset of electronic mail messages.

5. The method of claim 1, further comprising:

receiving, by the server, one or more properties indicating a request type of electronic mail messages to define a distribution list to which to assign a user associated with the client device; identifying, by the server, from a plurality of distribution lists, the distribution list to which to assign the user based on the one or more properties; selecting, by the server, from the plurality of electronic mail messages, a second subset of electronic mail messages based on assignment of the user to the distribution list and the request type determined from content in the second subset of electronic mail messages; and transmitting, by the server, to the client device, an identification of the second subset of electronic mail messages for presentation in the message panel.

6. The method of claim 1, further comprising:

receiving, by the server, from the client device, a request to change assignment of an electronic mail message of a second subset of electronic mail messages in the plurality of electronic mail messages to another user; identifying, by the server, from a plurality of users, a second user to which to assign the electronic mail message based on the request to change; and sending, by the server, to a second client device associated with the second user, an identification of the electronic mail message.

7. The method of claim 1, further comprising:

identifying, by the server, from a second subset of electronic mail messages in the plurality of electronic mail message, an electronic mail message for which no response is sent within a defined time window after receipt; and sending, by the server, to the client device, an indicator identifying the electronic mail message as having no response within the defined time window via the user interface.

8. A system, comprising:

at least one server having one or more processors coupled with memory in communication with a client device, the server configured to:

receive, from the client device, a request to present an electronic mail thread corresponding to a subset of electronic mail messages in a plurality of electronic mail messages in a first message panel of a user interface in an application, as a chat conversation thread in a second message panel of the interface; identify, from each of the subset of electronic mail messages, main content and auxiliary content, responsive to the request to present; generate the chat conversation thread including the main content identified from each of the subset of electronic mail messages, while excluding the auxiliary content; send, to the client device, the chat conversation thread to be presented in the second message panel of the user interface in the application and to remove the subset of electronic mail messages from presentation in the first message panel; receive, from the client device, an input on the second message panel corresponding to a reply message to at least one of the subset of electronic mail messages; and

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cause the client device to present the reply message as part of the chat conversation thread on the second message panel.

9. The system of claim 8, wherein the at least one server is further configured to:

receive, from the client device, a request to add a second chat conversation thread as one of the plurality of electronic mail messages in the first message panel; generate an electronic mail thread to include a second subset of electronic mail messages corresponding to a plurality of conversations of the second chat conversation thread; and

send, to the client device, an identification of the electronic mail thread among a plurality of threads for presentation the message panel.

10. The system of claim 8, wherein the at least one server is further configured to generate the chat conversion thread in accordance with a format defining presentation of the chat conversation thread in the message panel.

11. The system of claim 8, wherein the at least one server is further configured to identify, from a second application separate from the application, a second chat conversation to be converted to a second subset of electronic mail messages.

12. The system of claim 8, wherein the at least one server is further configured to:

receive one or more properties indicating a request type of electronic mail messages to define a distribution list to which to assign a user associated with the client device; identify, from a plurality of distribution lists, the distribution list to which to assign the user based on the one or more properties;

select, from the plurality of electronic mail messages, a second subset of electronic mail messages based on assignment of the user to the distribution list and the request type determined from content in the second subset of electronic mail messages; and

transmit, to the client device, an identification of the second subset of electronic mail messages for presentation in the message panel.

13. The system of claim 8, wherein the at least one server is further configured to:

receive, from the client device, a request to change assignment of an electronic mail message of a second subset of electronic mail messages in the plurality of electronic mail messages to another user;

identify, from a plurality of users, a second user to which to assign the electronic mail message based on the request to change; and

send, to a second client device associated with the second user, an identification of the electronic mail message.

14. The system of claim 8, wherein the server is distributed in a cloud computing network in communication with a plurality of client devices including the client device.

15. A system, comprising:

a client device having one or more processors coupled with memory in communication with a server; and an application executable on the client device, the application having a user interface comprising a first message panel and a second message panel, the application configured to:

receive, via the user interface, a request to present an electronic mail thread corresponding to a subset of electronic mail messages in a plurality of electronic

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mail messages in the first message panel in the user interface, as a chat conversation thread in the second message panel;

identify, from each of the subset of electronic mail messages, main content and auxiliary content, responsive to the request to present;

generate the chat conversation thread including the main content identified from each of the subset of electronic mail messages, while excluding the auxiliary content;

present, in the second message panel of the user interface, at least a portion of the chat conversation thread;

remove, from presentation in the first message panel of the user interface, the subset of electronic messages;

receive, via the second message panel, an input corresponding to a reply message to at least one of the subset of electronic mail messages; and

present the reply message as part of the chat conversation thread on the second message panel.

16. The system of claim 15, wherein the application is further configured to generate the chat conversion thread in accordance with a format defining presentation of the chat conversation thread in the message panel.

17. The system of claim 15, wherein the application is further configured to identify, from a second application separate from the application, a second chat conversation thread to be converted to a second subset of electronic mail messages.

18. The system of claim 15, wherein the application is further configured to:

identify, via the user interface, one or more properties indicating a request type of electronic mail messages to define a distribution list to which to assign a user of the client device;

transmit, to the server, the one or more properties to assign the user of the client device to the distribution list, with which at least one of the plurality of electronic mail messages; and

receive, from the server, an identification of a second subset of electronic mail messages selected from the plurality of electronic mail messages based on an assignment of the user of the client device to the distribution list and the request type determined from content in the second subset of electronic mail messages.

19. The system of claim 15, wherein the application is further configured to:

identify, from the plurality of electronic mail messages, an electronic mail message for which no response is sent within a defined time window after receipt; and

provide, via the user interface, an indicator identifying the electronic mail message as having no response within the defined time window.

20. The method of claim 15, wherein the client device comprises a mobile communication device in communication with the server on a cloud computing network, wherein the user interface has a size corresponding to a display of the mobile communication device.

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